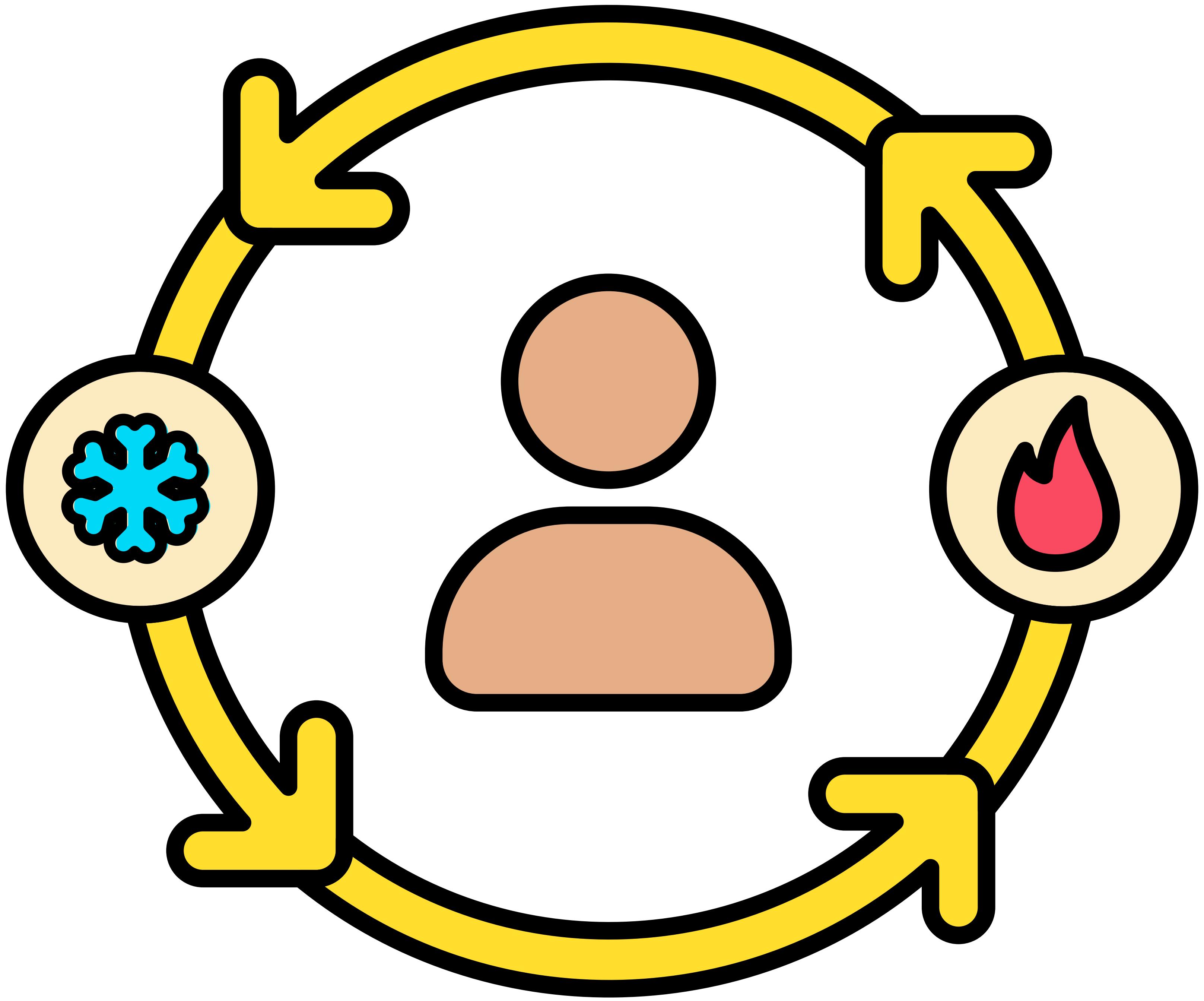


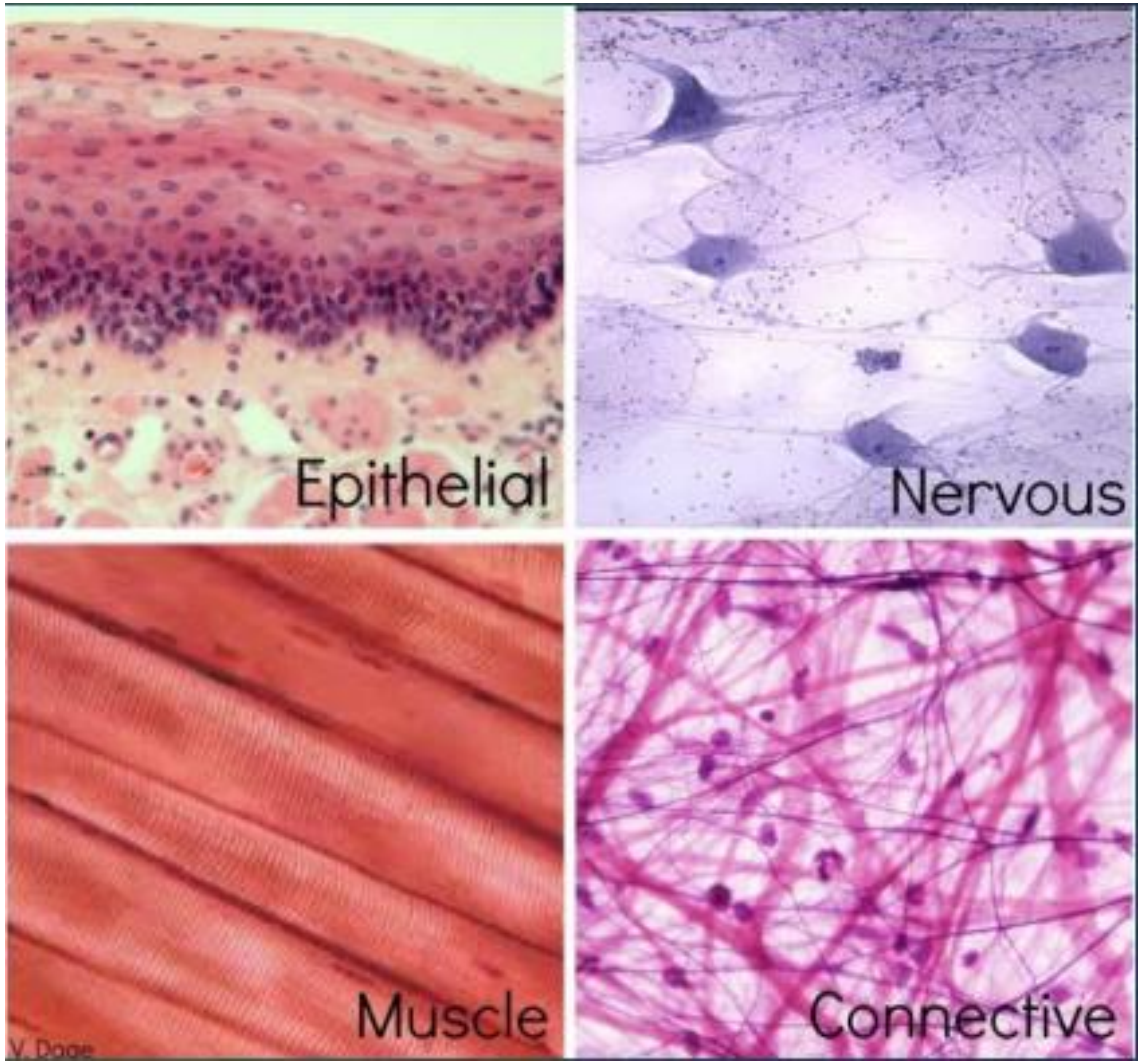
Fundamentals of Homeostasis

Content

1. Explain the basic organization of the body
2. Define the fluid compartments of the body
3. Explain homeostasis
4. Explain homeostasis control
5. Explain circadian rhythms
6. Explain how solutes distribute within the body
7. Contrast how the transporters, pumps, and channels work
8. Describe how the ion channels are gated

Body components

- 1 fertilized egg ⟹ division + differentiation
- Diferentiated cells = specialized function
- Tissues = groups of cells with related function



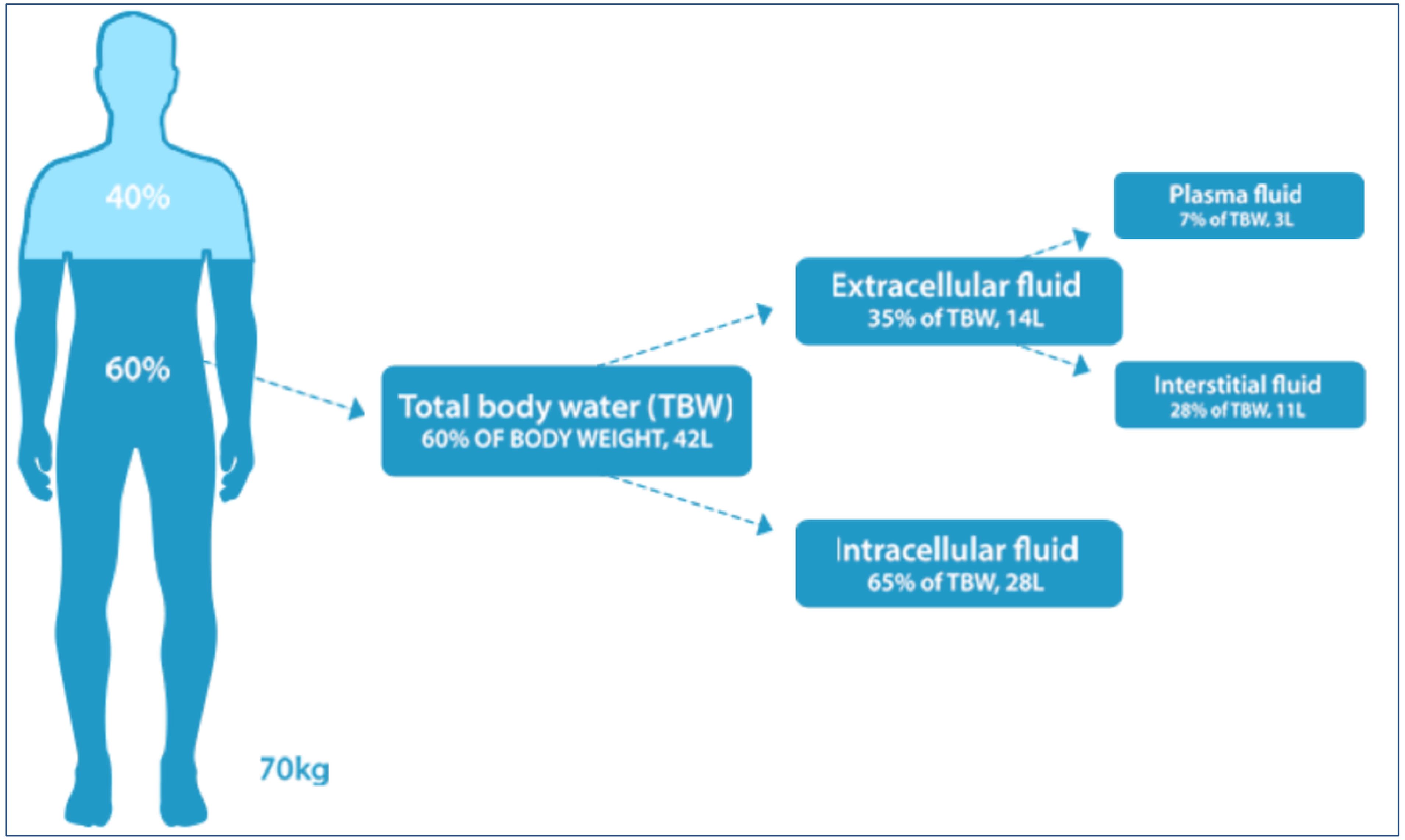
Body components

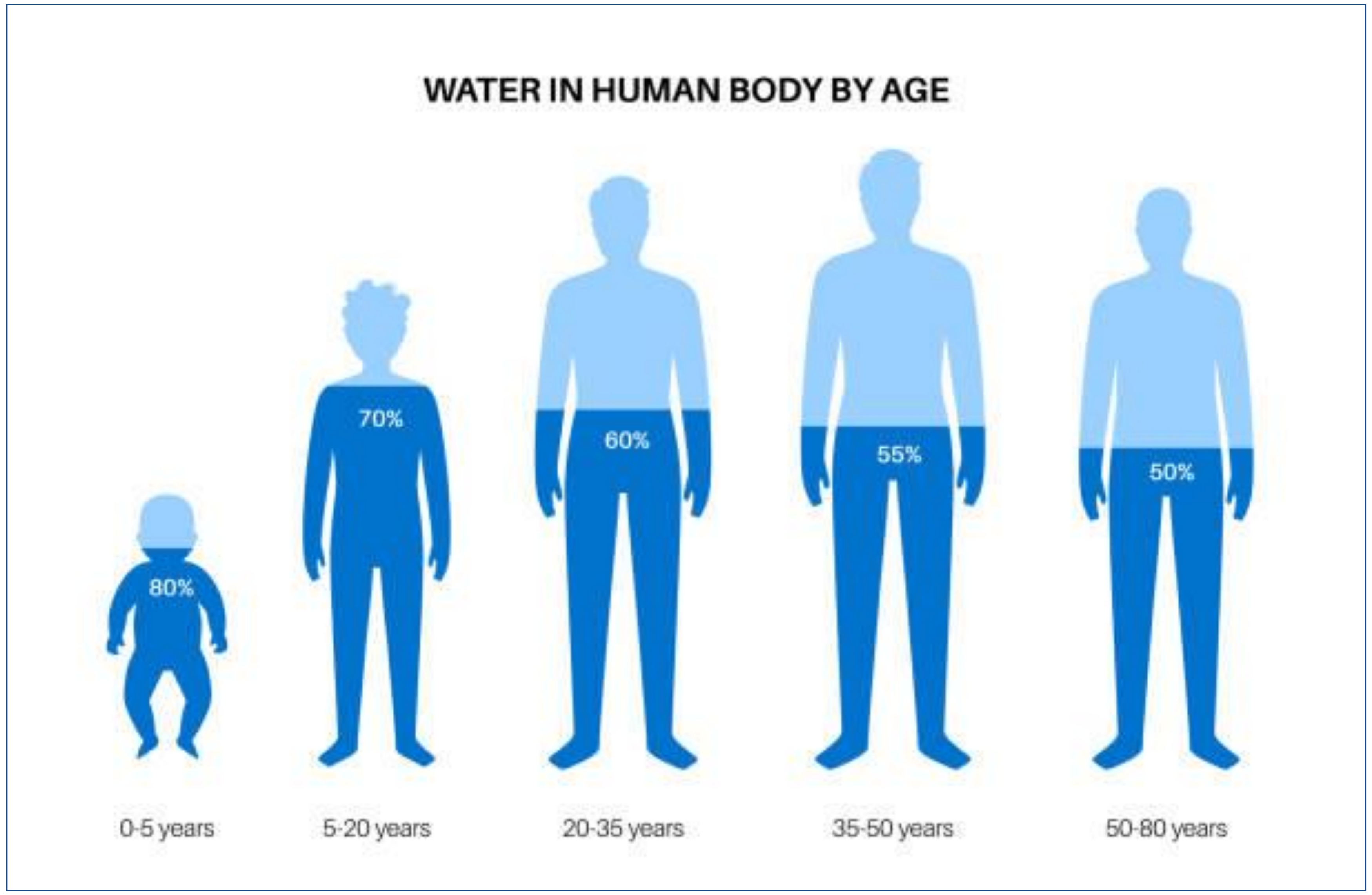
- Organ = function unit
- Organ system = several organs work together to perform specific function
- Skin = Barrier
- Entry = respiratory + Gastrointestinal tract
- Transport = Cardiovascular system
- Exit = Renal + Gastrointestinal tract

Body Fluid Compartments

Total human water = 60% Body weight

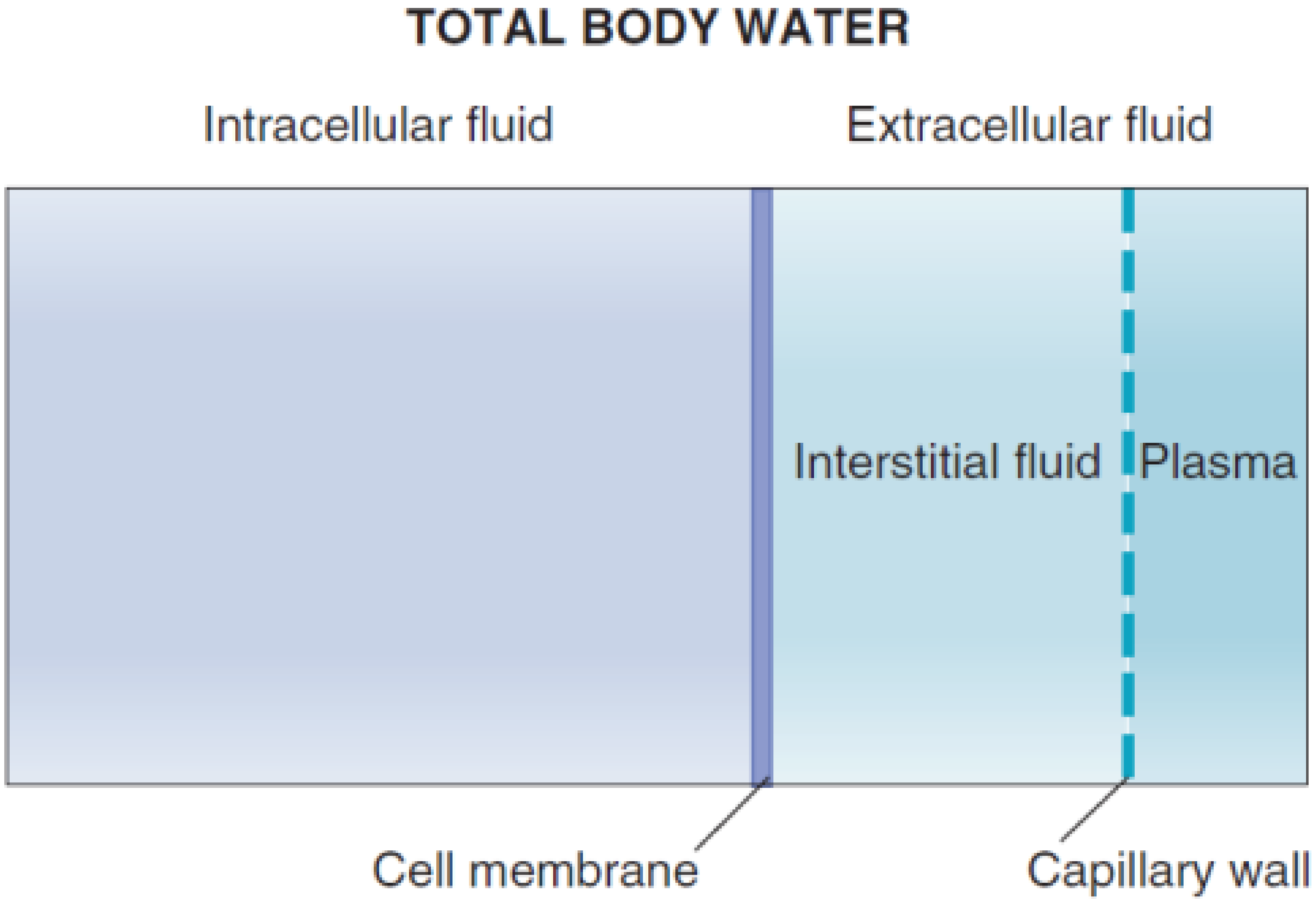
Total body weight correlates Inversely with body fat





Body Fluid Compartments

- ICF = 2/3 TBW
- ECF = 1/3 TBW
- ISF = 3/4 ECF
- Plasma = 1/4 ECF
- ISF = ultrafiltrate of plasma



Self-regulating mechanism

1. Equilibrium: IVF & ISF

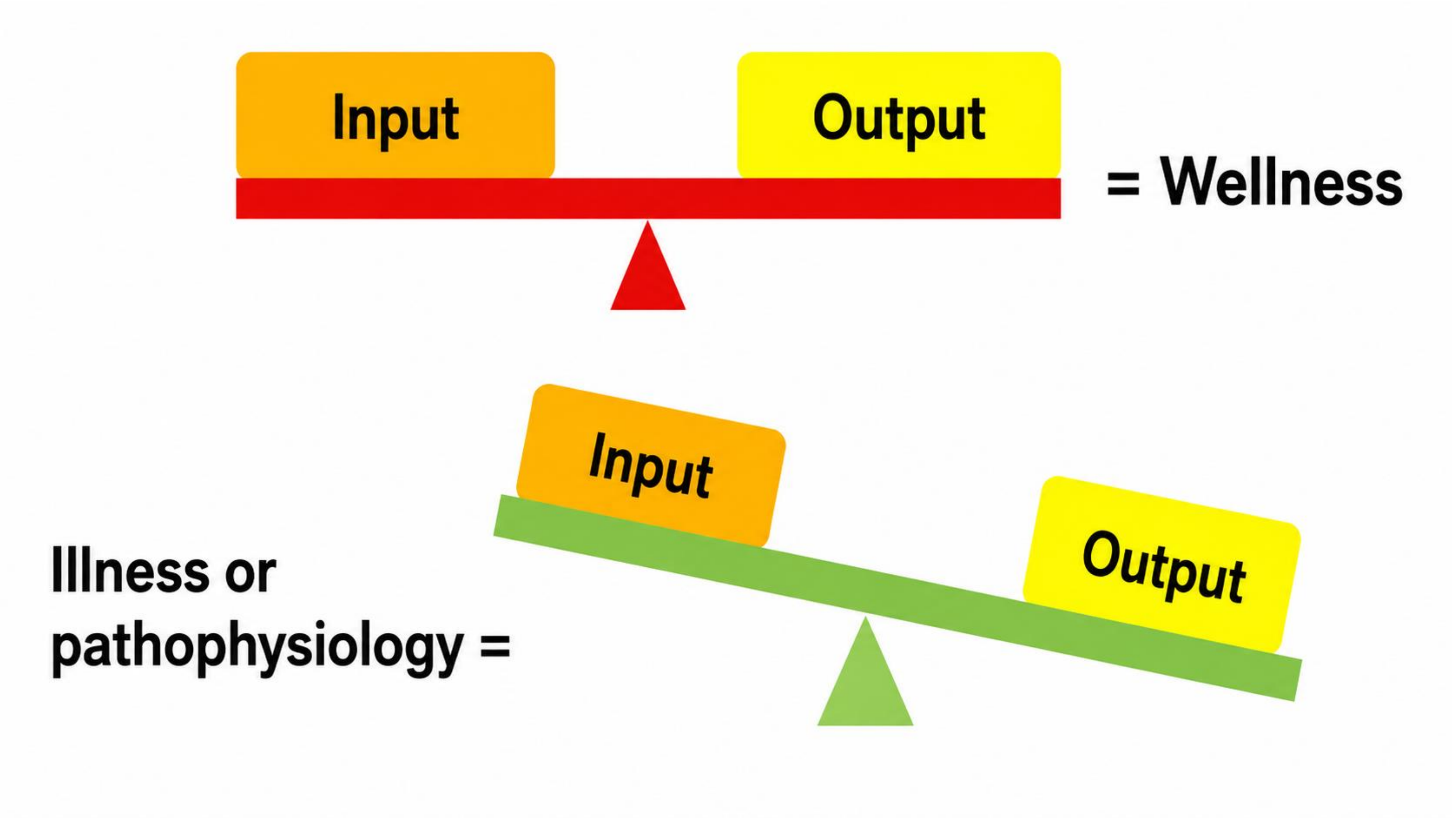
- Equal amount of substance
- No net transfer of substance or energy
- No barrier to movement
- No energy expenditure to maintain

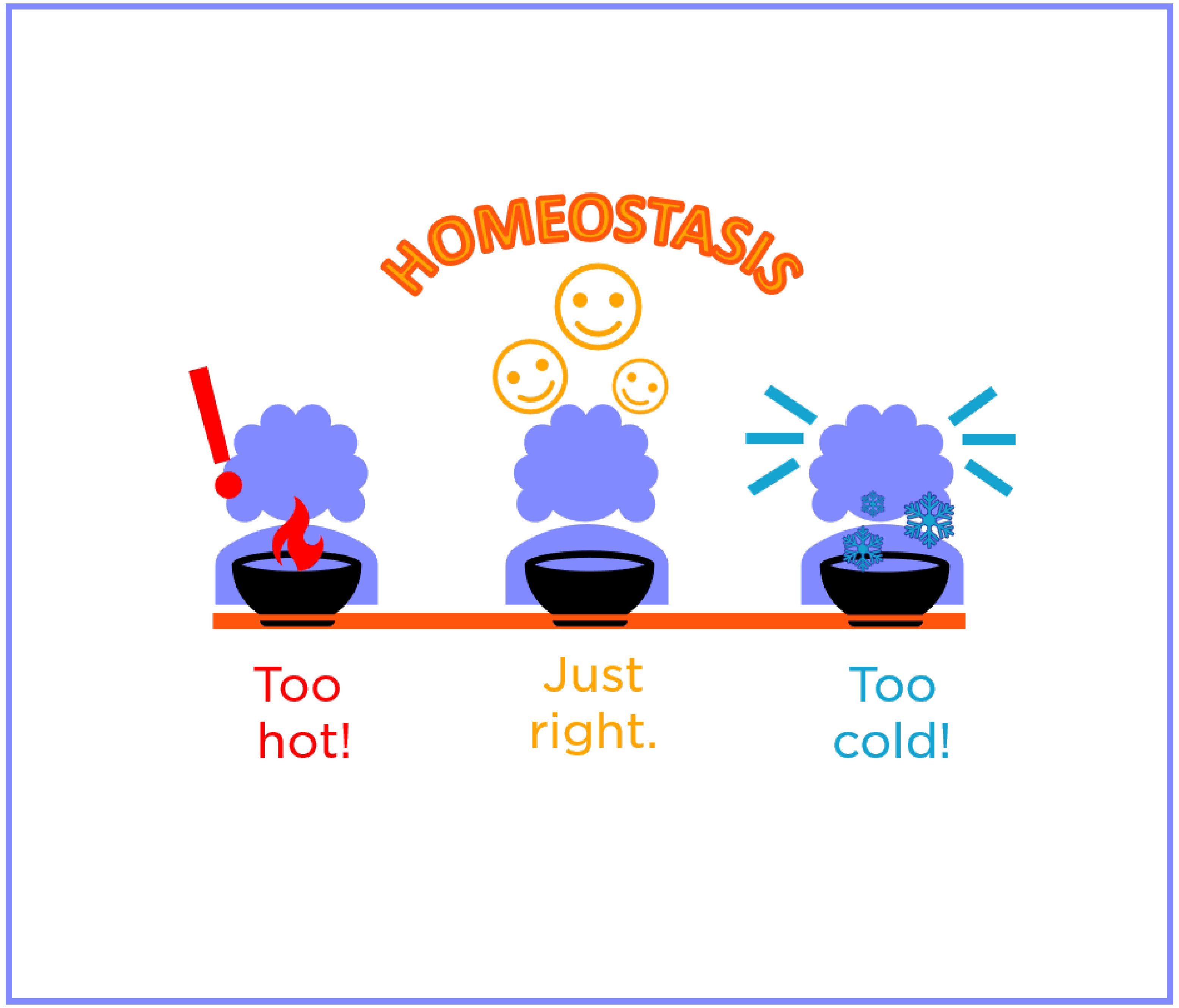
2. Steady state: ICF & ECF

- Constant amount of substance in compartments
- Input = output
- Require energy to maintain

What is homeostasis?

- ✓ Maintanence of ECF constituents as relatively constant
- ✓ Central theme of physiology





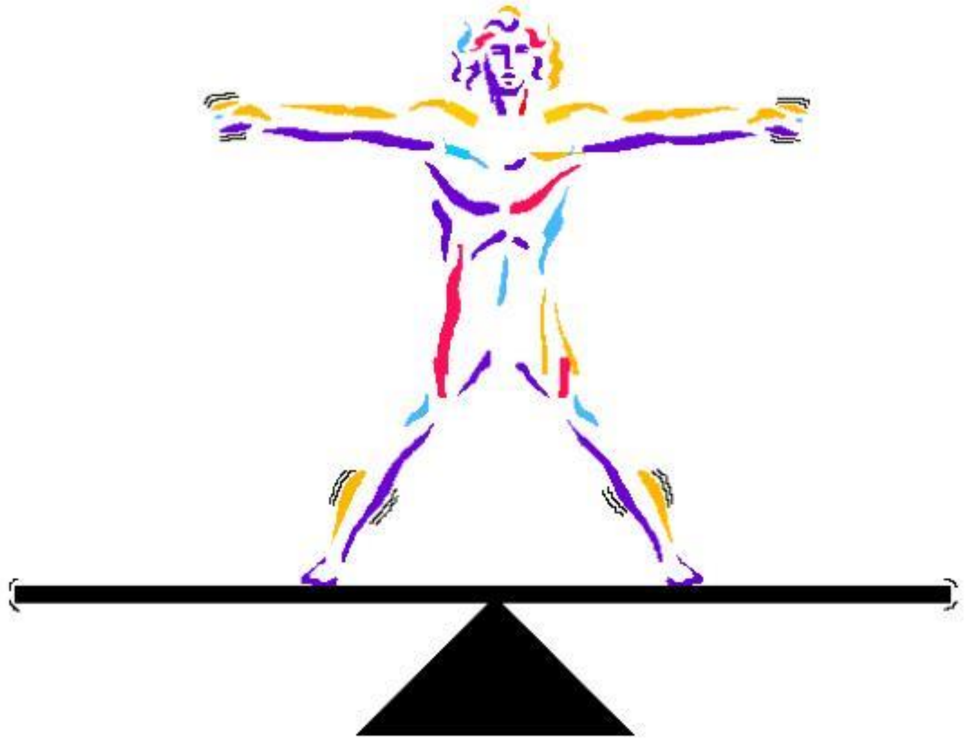
Homeostasis

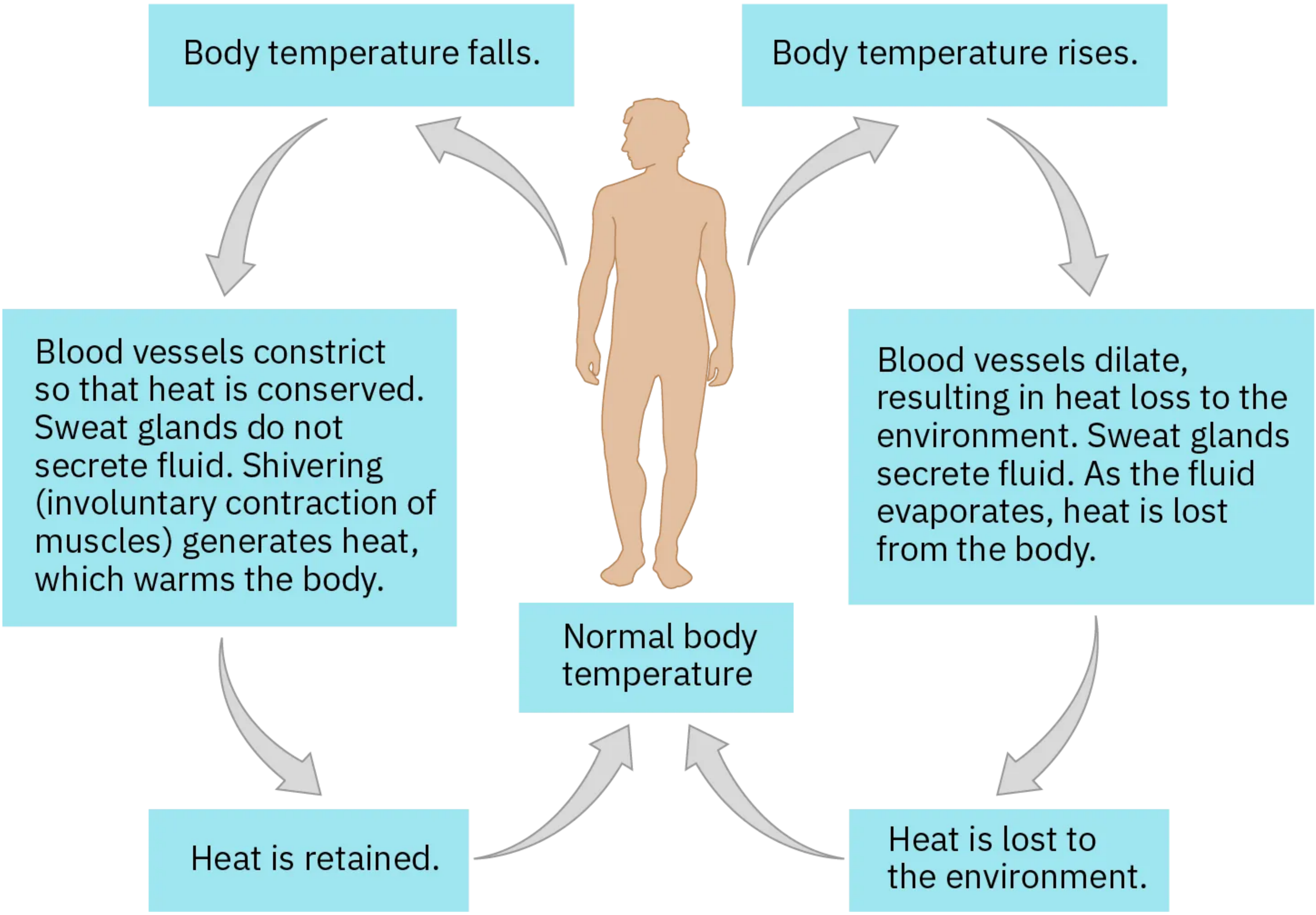
Examples of Homeostasis

Many variables are maintained by homeostasis.

Examples include:

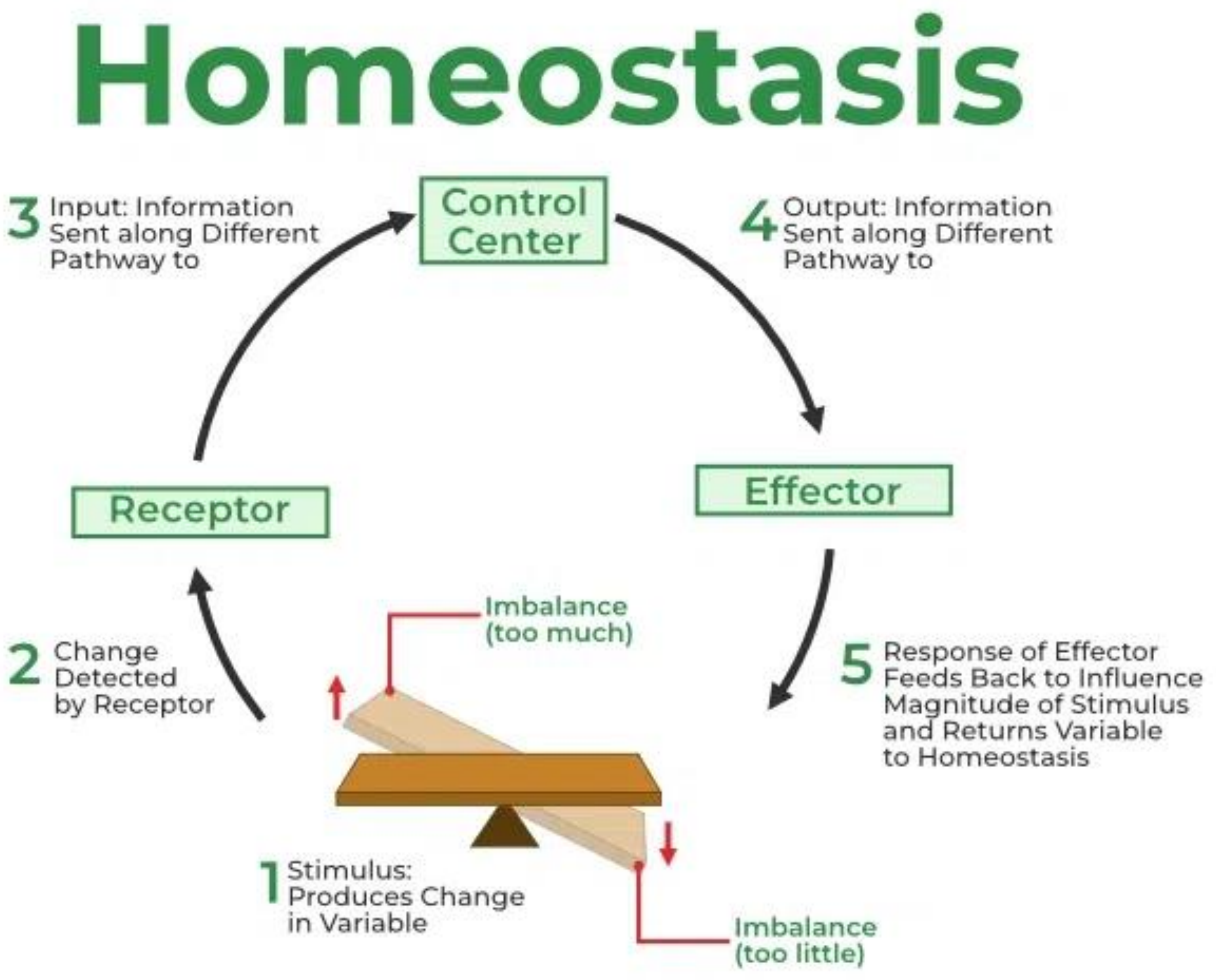
* Temperature
* Blood pH
* Blood sugar
* Water balance
* Blood pressure
* Ion balance



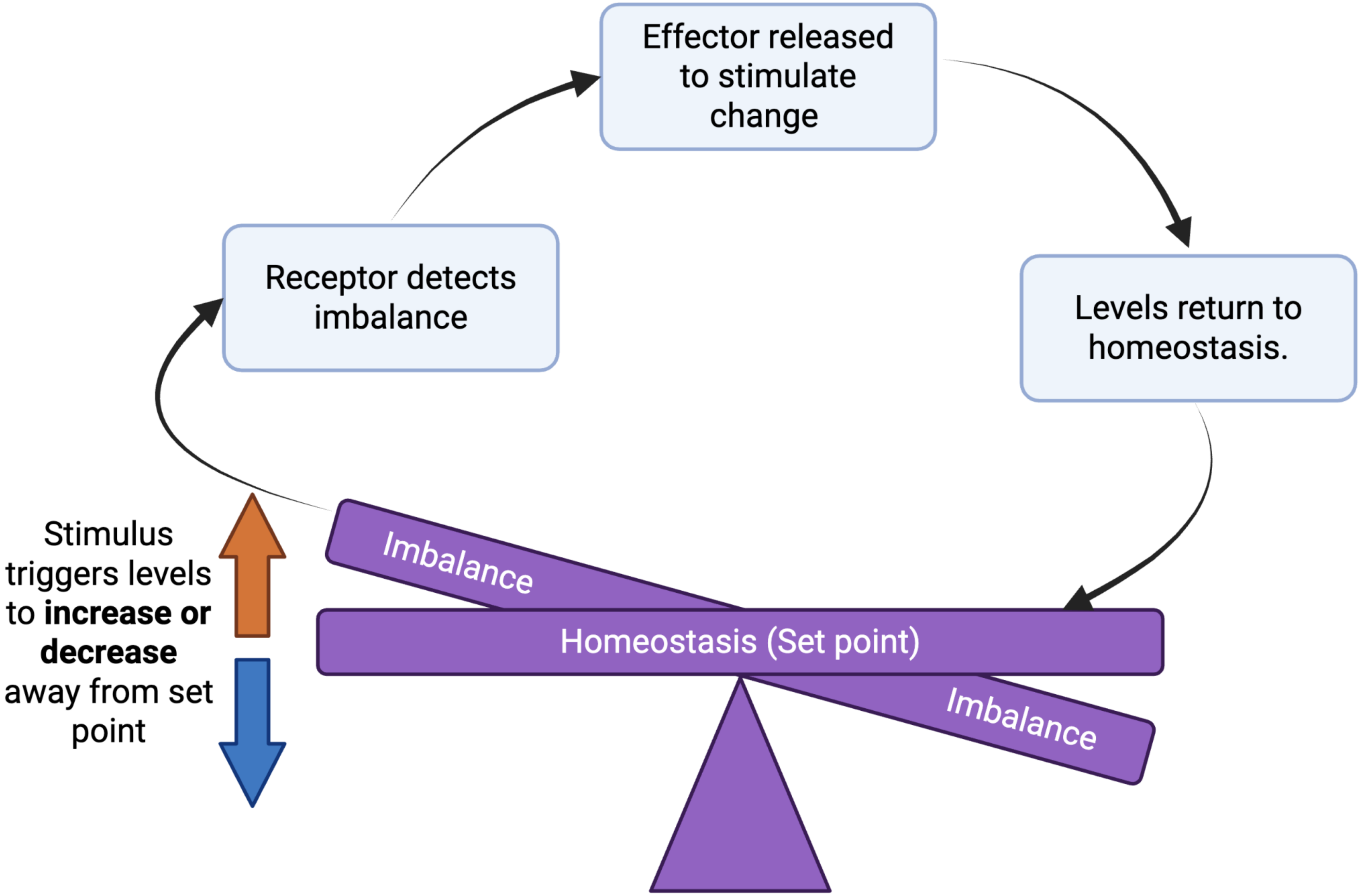


Homeostasis Control Mechanism

Homeostasis is maintained through a continuous **feedback loop** that detects internal imbalances and initiates corrective responses to restore equilibrium.



Feedback loop



Stimulus

A change in an environmental variable

Sensor/Receptor

Monitors and detects changes

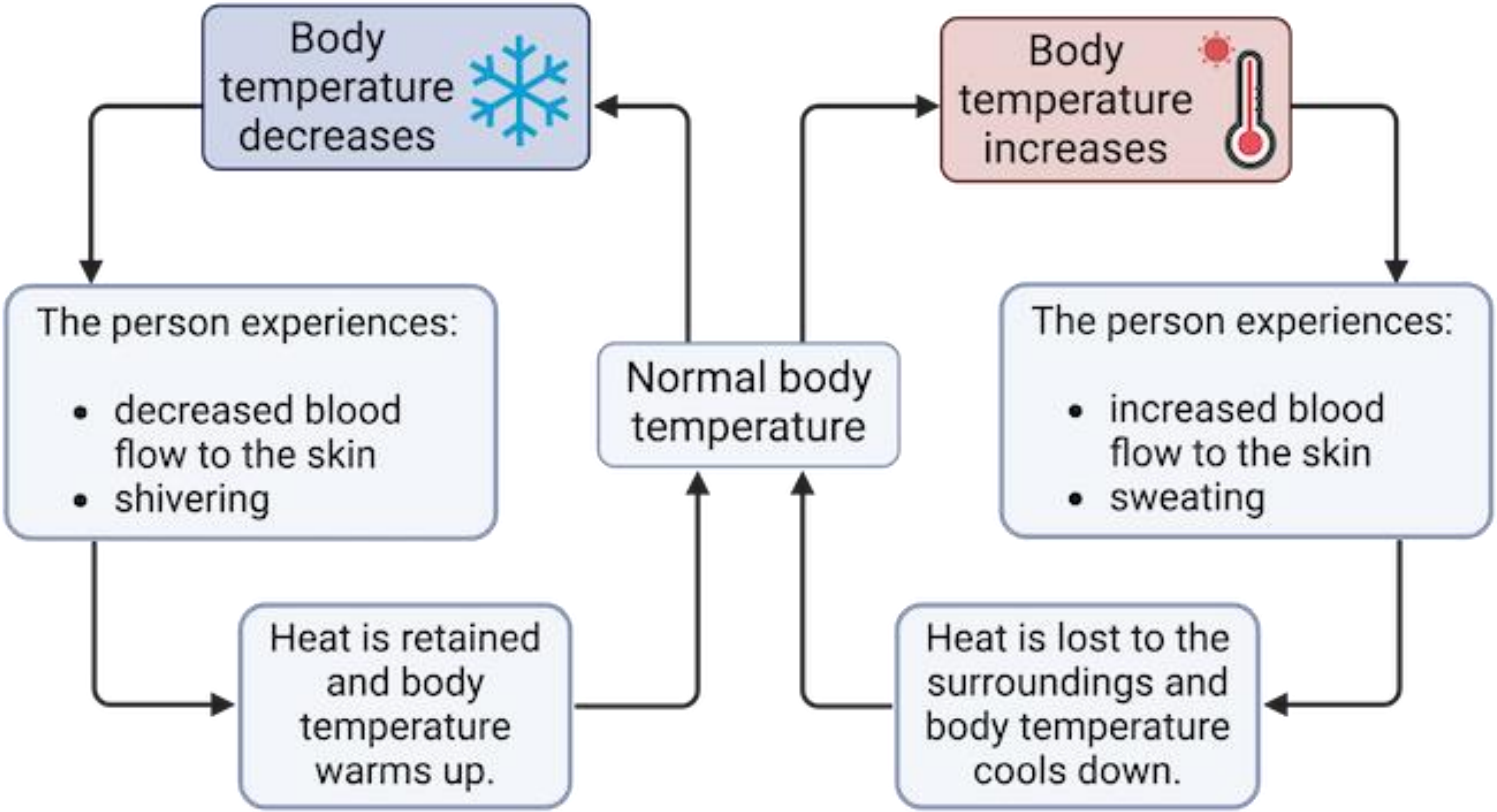
Control Center

Analyzes and compares with the "Set point".

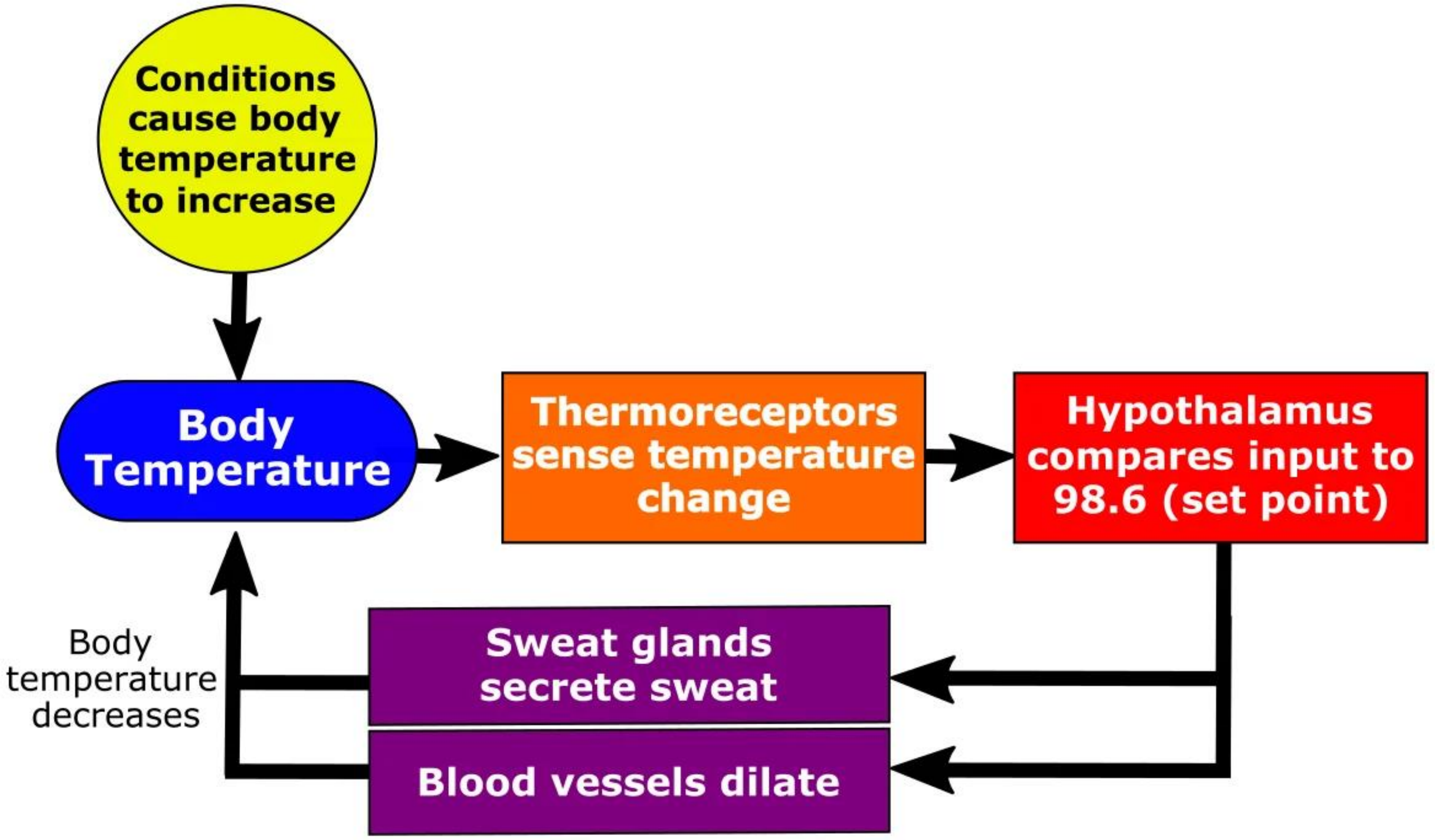
Effector

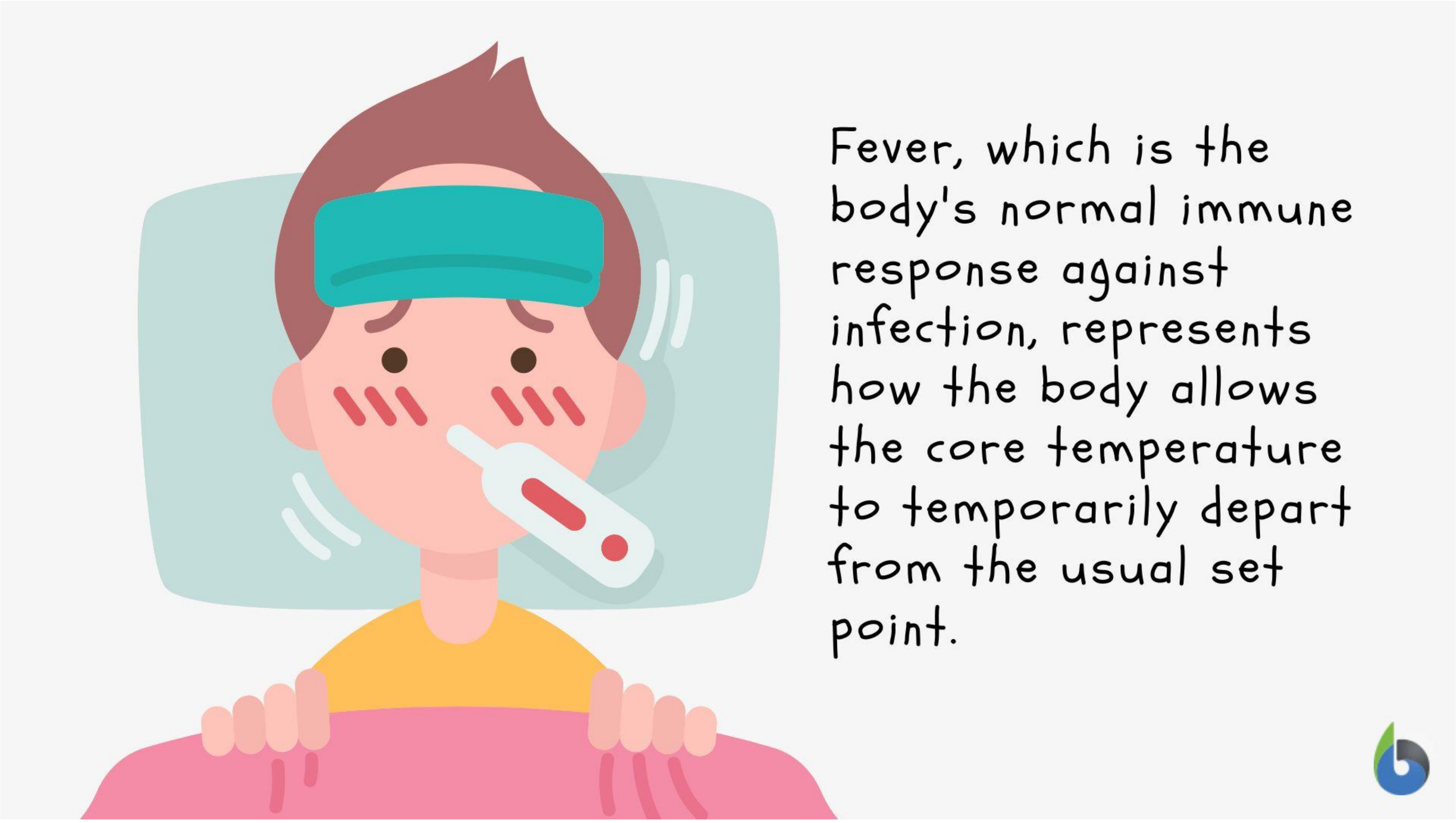
Executes corrective responses

Temperature reflex loops: external change

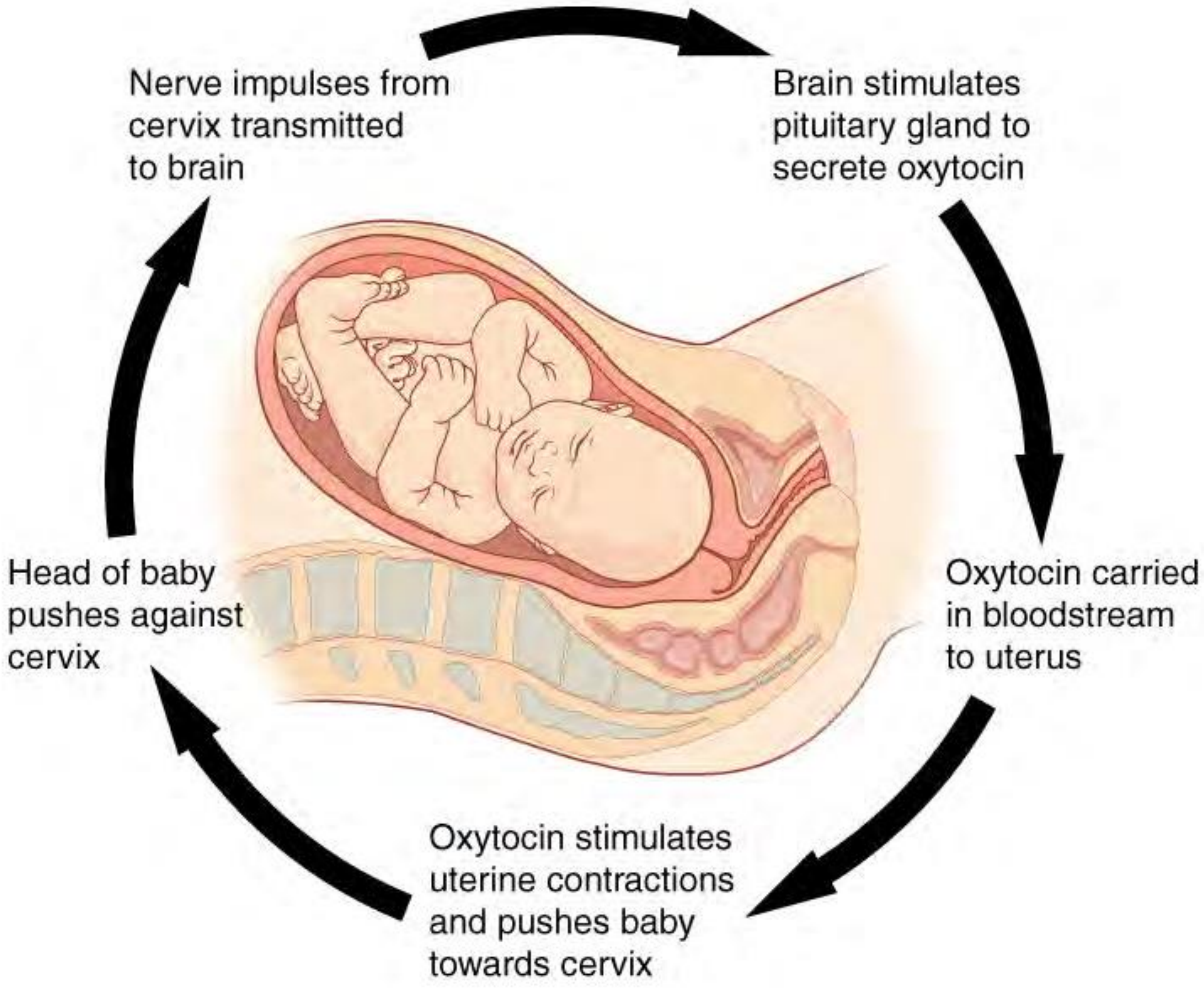


Temperature reflex loops: external change





Possotive Feedback in Childbirth



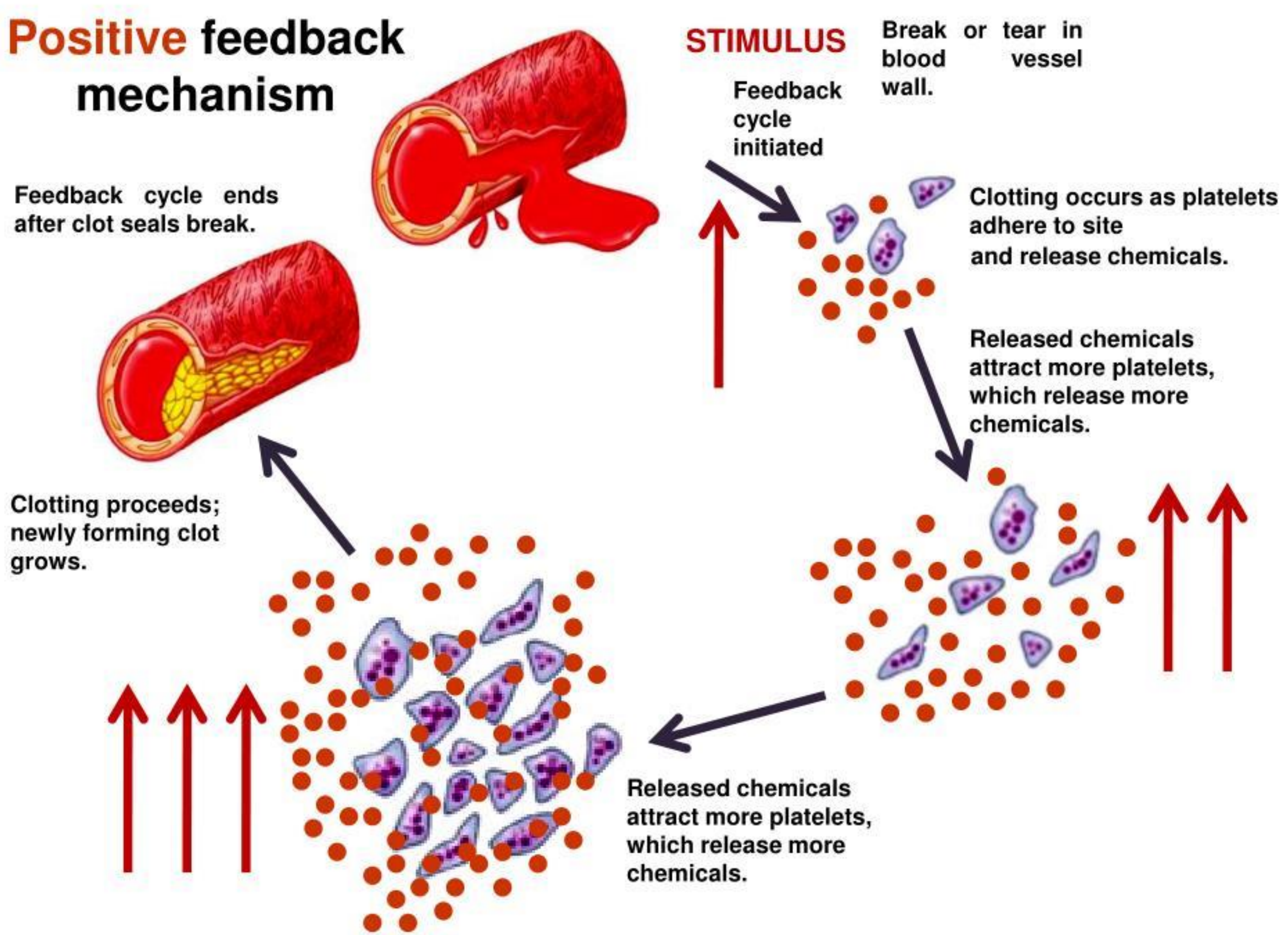
The Oxytocin Cascade: Self-reinforcing hormonal release until childbirth is complete

Amplification Response enhances the original stimulus.

Fixed Endpoint Drives system to a specific conclusion.

Dynamic Cycle Pressure → Signal → Hormone → Force.

Possitive Feedback in Blood Clotting



The Signal Cascade Injury triggers platelet adhesion ⟶ chemicals (ADP and Thromboxane A2 (TXA2))

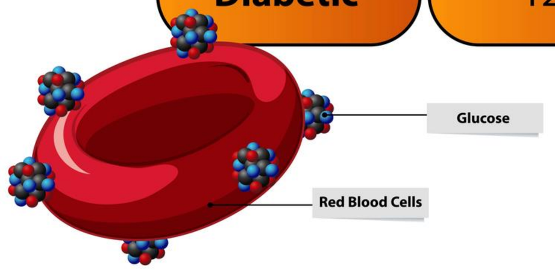
Self-Amplification These chemical signals act as chemo-attractants, recruiting a surge of nearby platelets to the site.

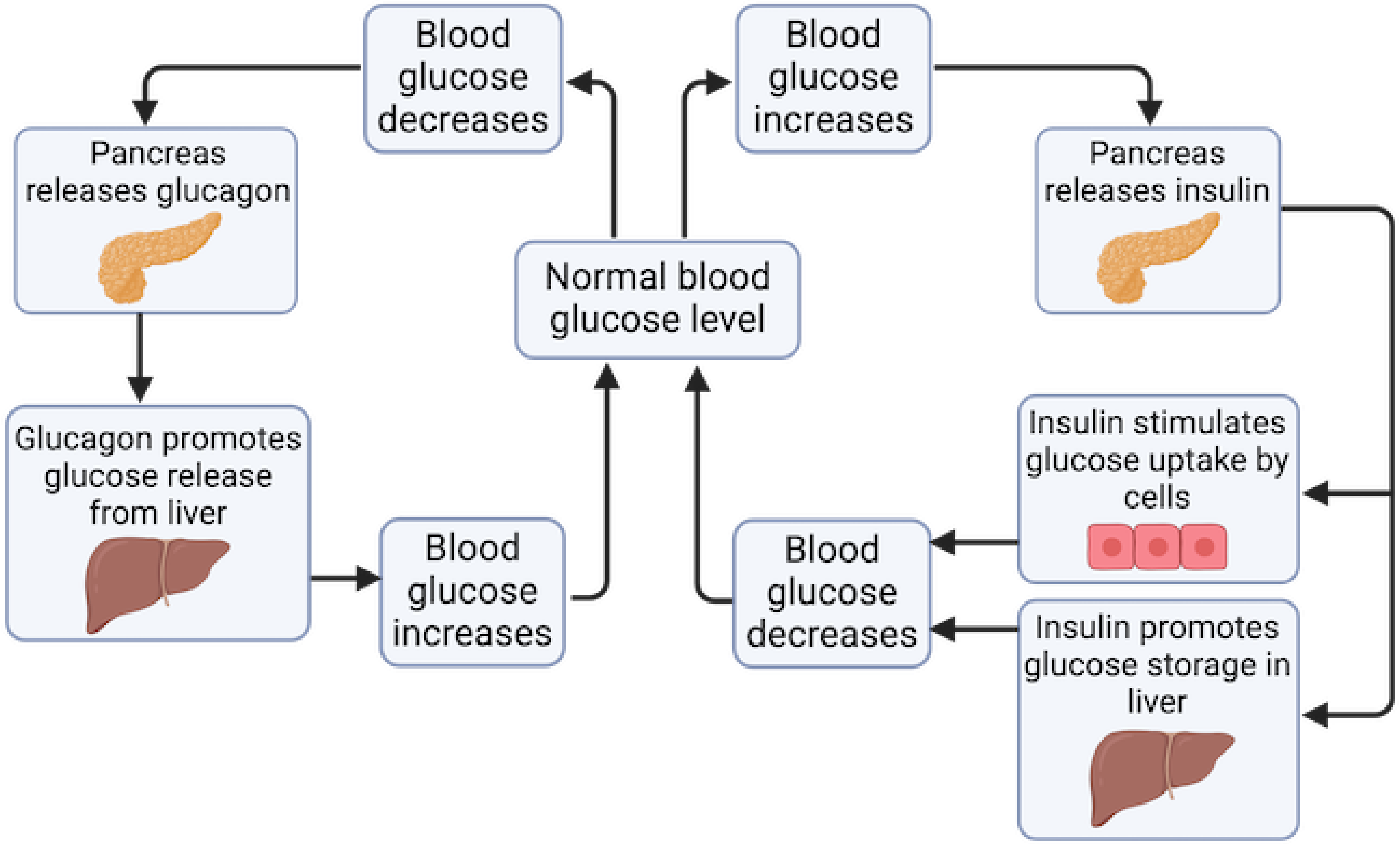
Completion Clot-sealed break

Negative Feedback

Homeostasis and Blood glucose level

Blood Glucose Chart			
Mg/DL	Fasting	After Eating	2-3 Hours After Eating
Normal	80-100	170-200	120-140
Impaired Glucose	101-125	190-230	140-160
Diabetic	126+	220-300	200+





Diabetes

- Type 1
- Type 2

Negative Feedback

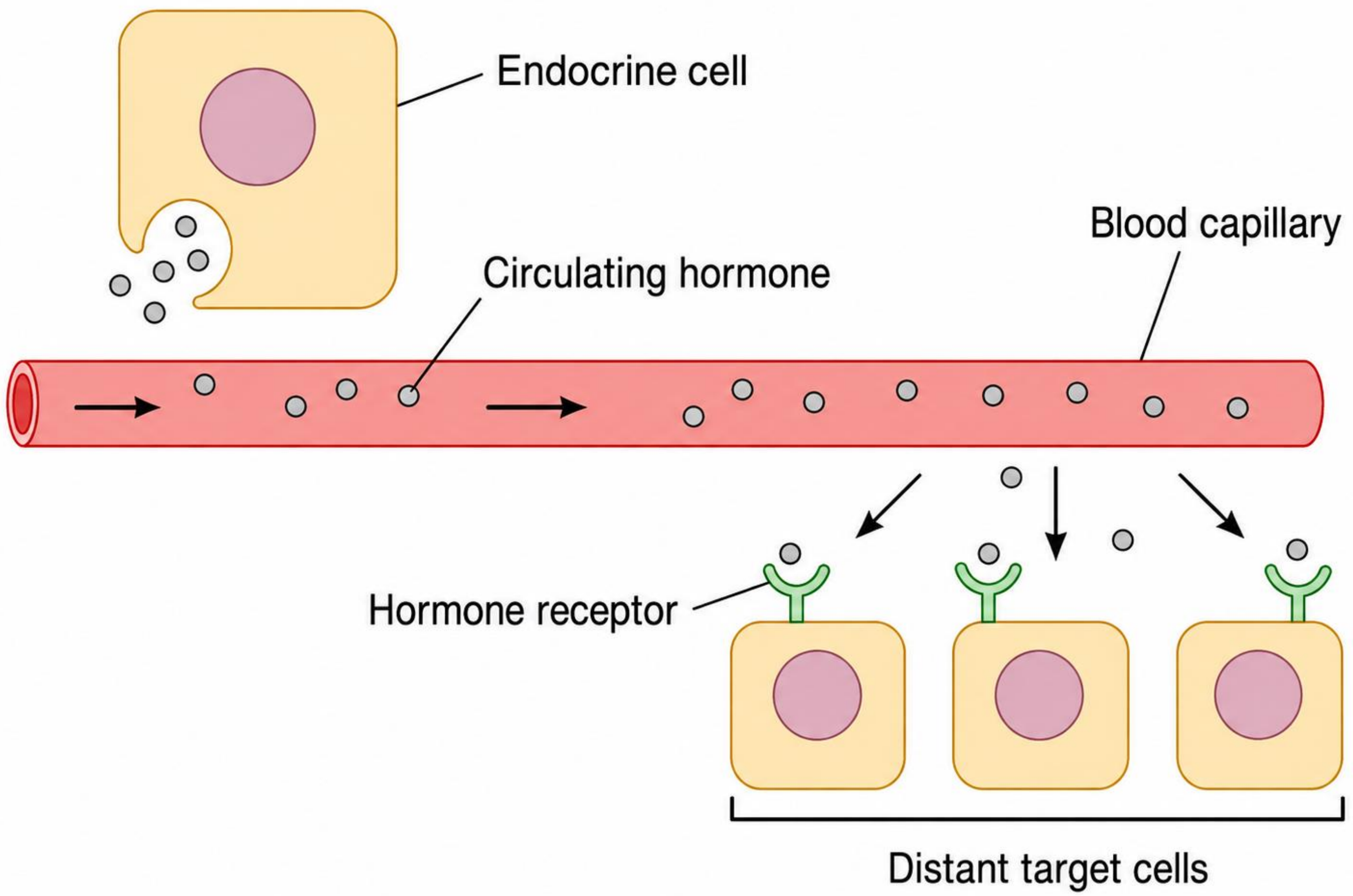
A mechanism that prevents a physiological response from going beyond the normal range by reversing the action once the normal range is exceeded

Positive Feedback

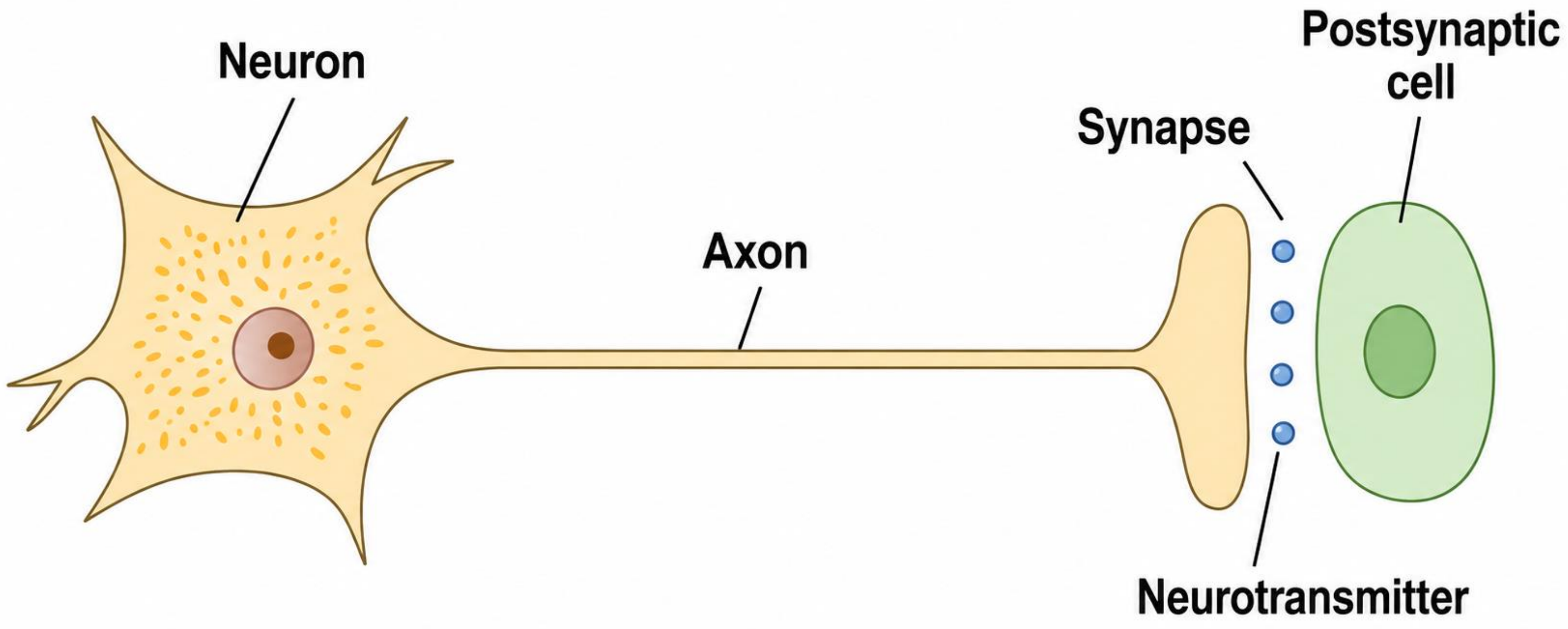
Intensifies a change in the body's physiological condition

Homeostasis control

Reflexes = response made at a distance from target cells

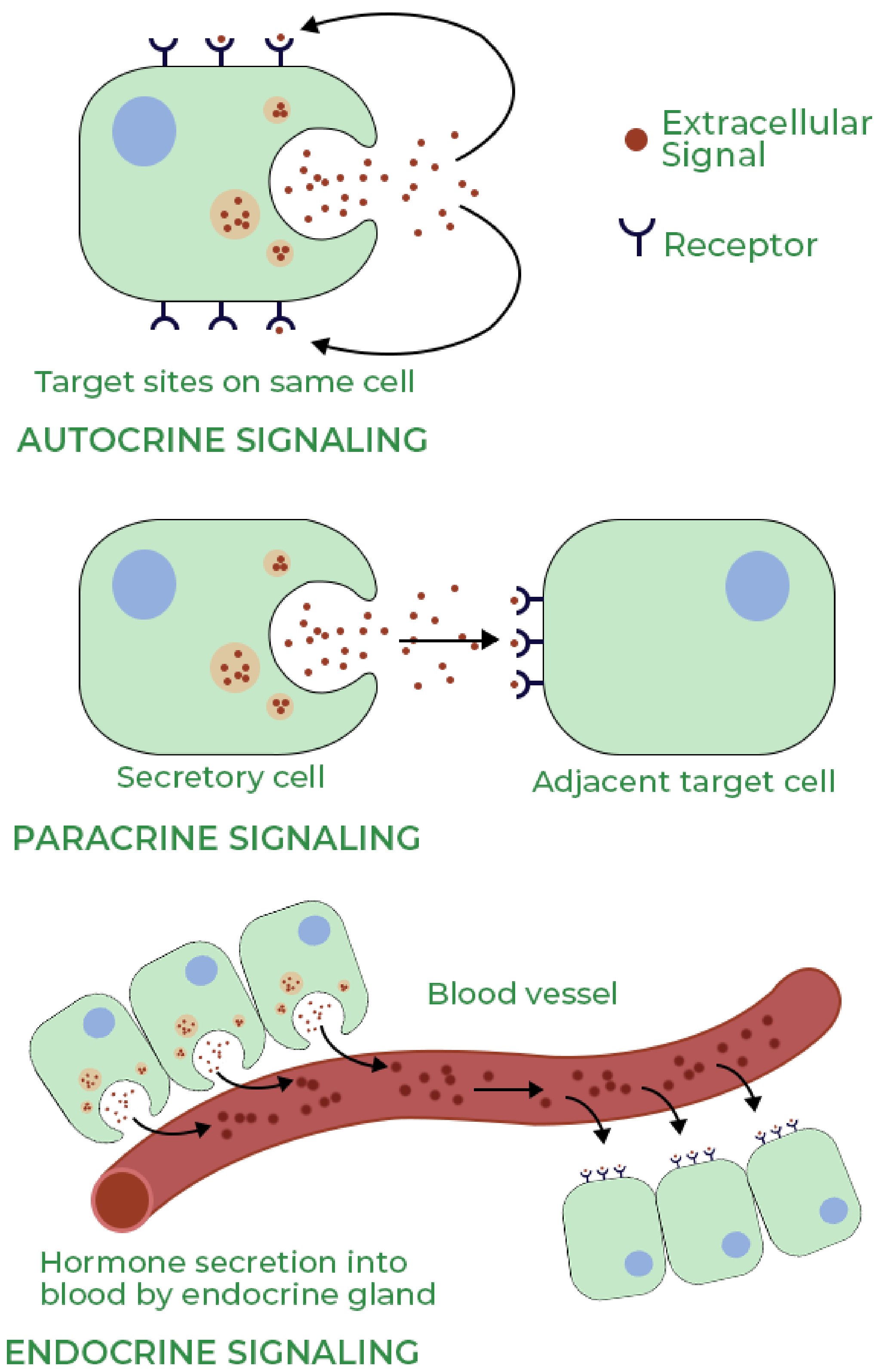


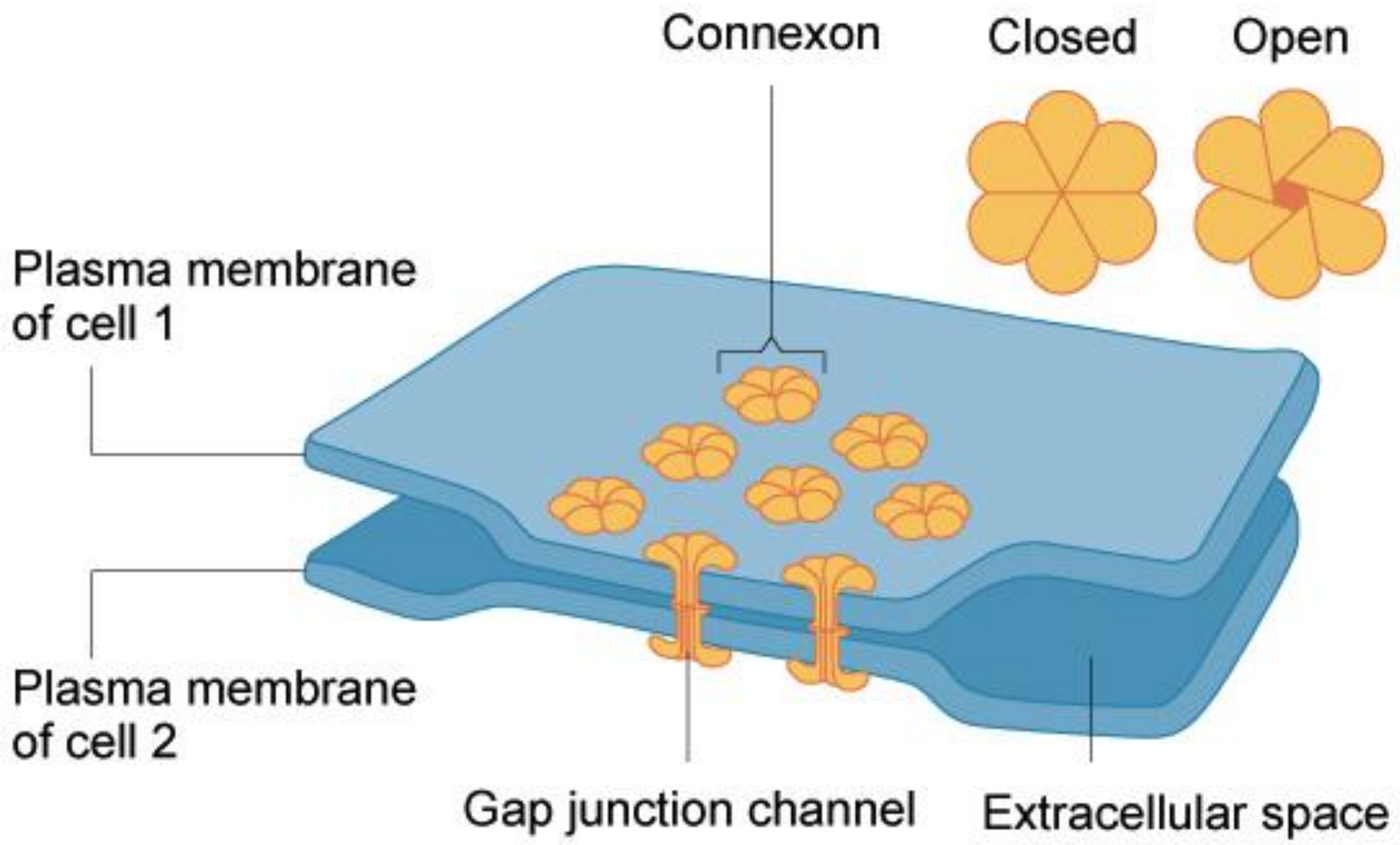
Neural



Homeostasis control

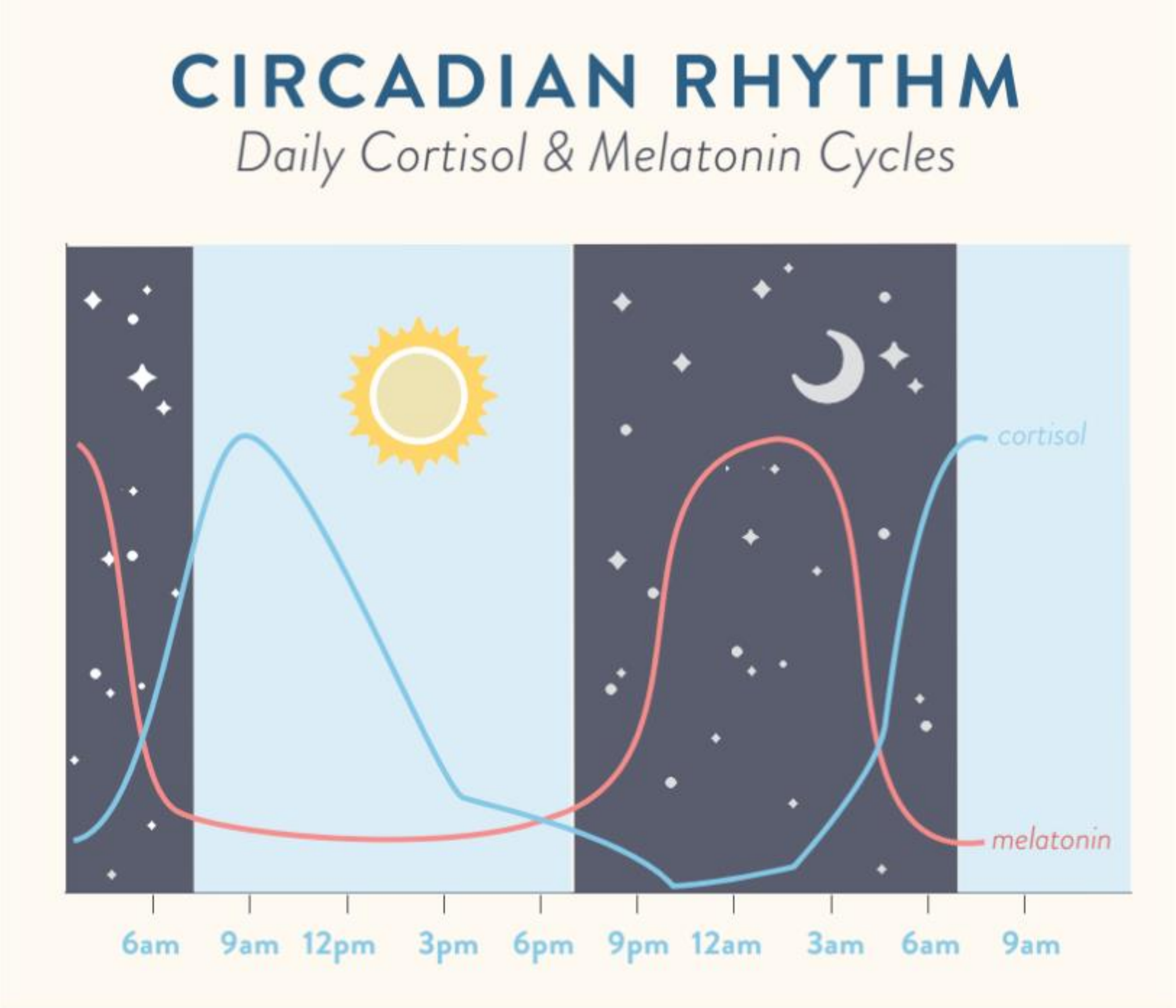
Local responses = occur at target cells: paracrine, autocrine, gap junction

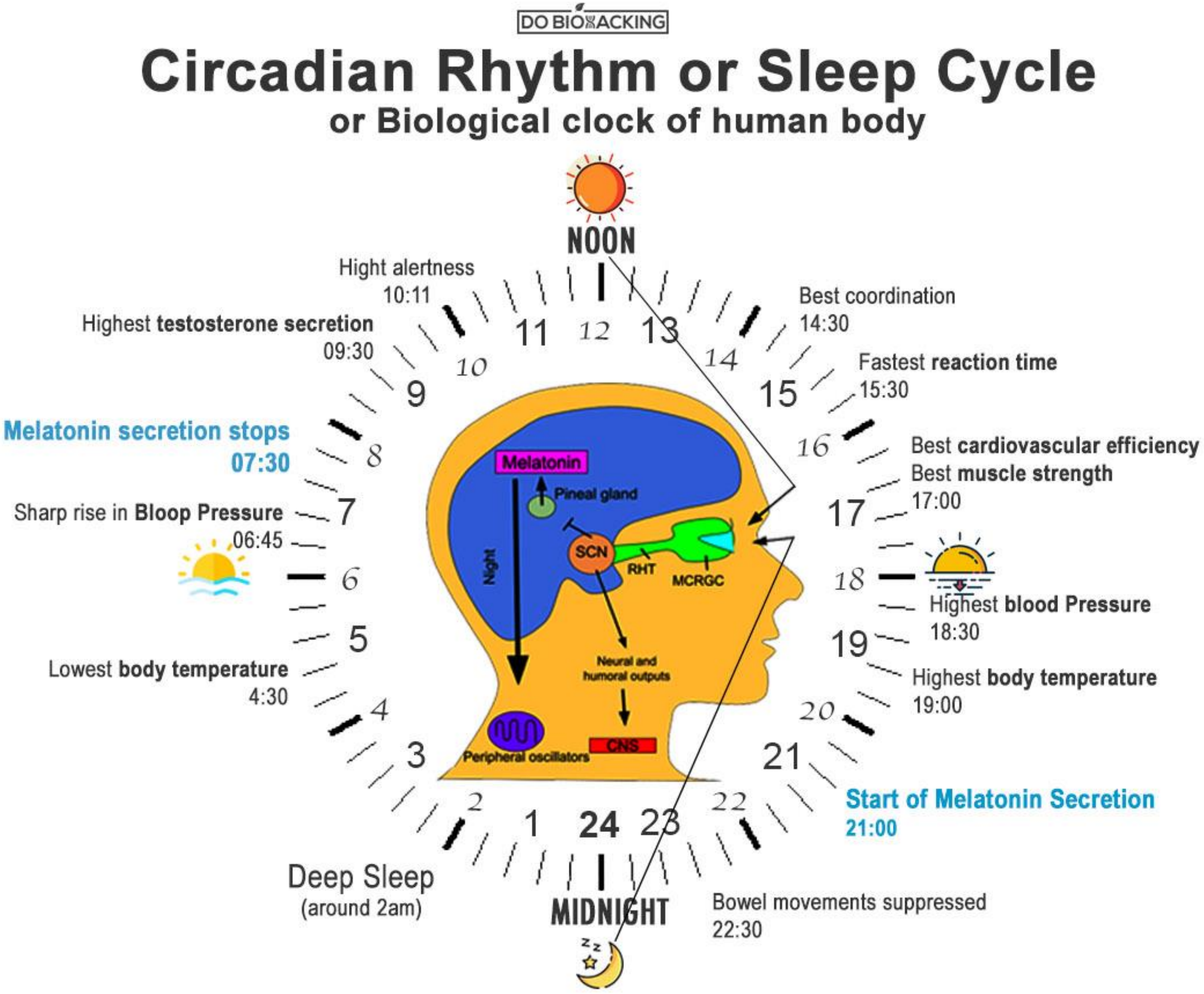




Rhythm

- 24h= circadian rhythm: Cortisol, GH, aldosterone, testosterone
- >24h = infradian rhythm: menstrual cycle, thyroid hormone increasing in winter.
- < 24h = ultradian rhythm: LH, prolactin.

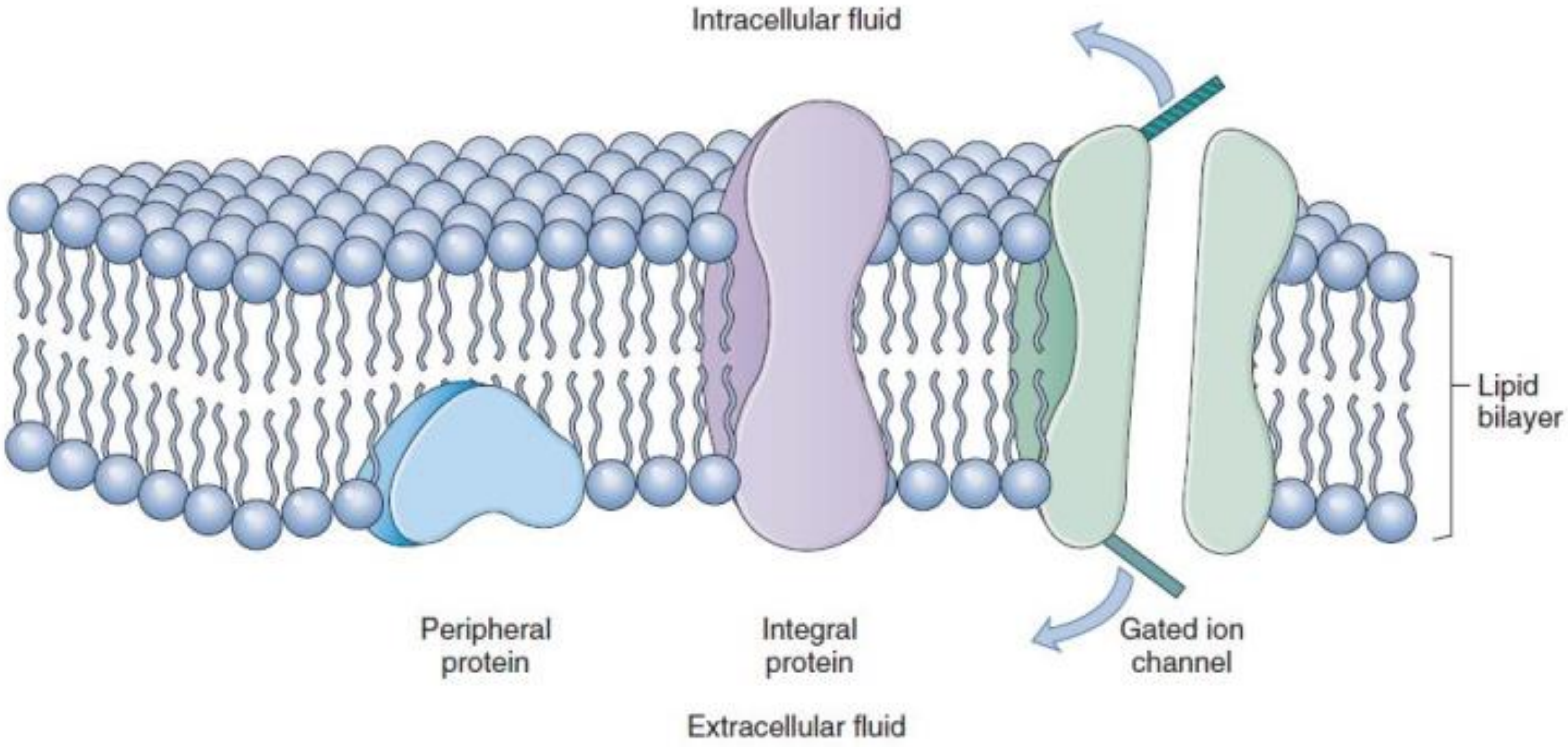




Importance of pumps, transporters & channels

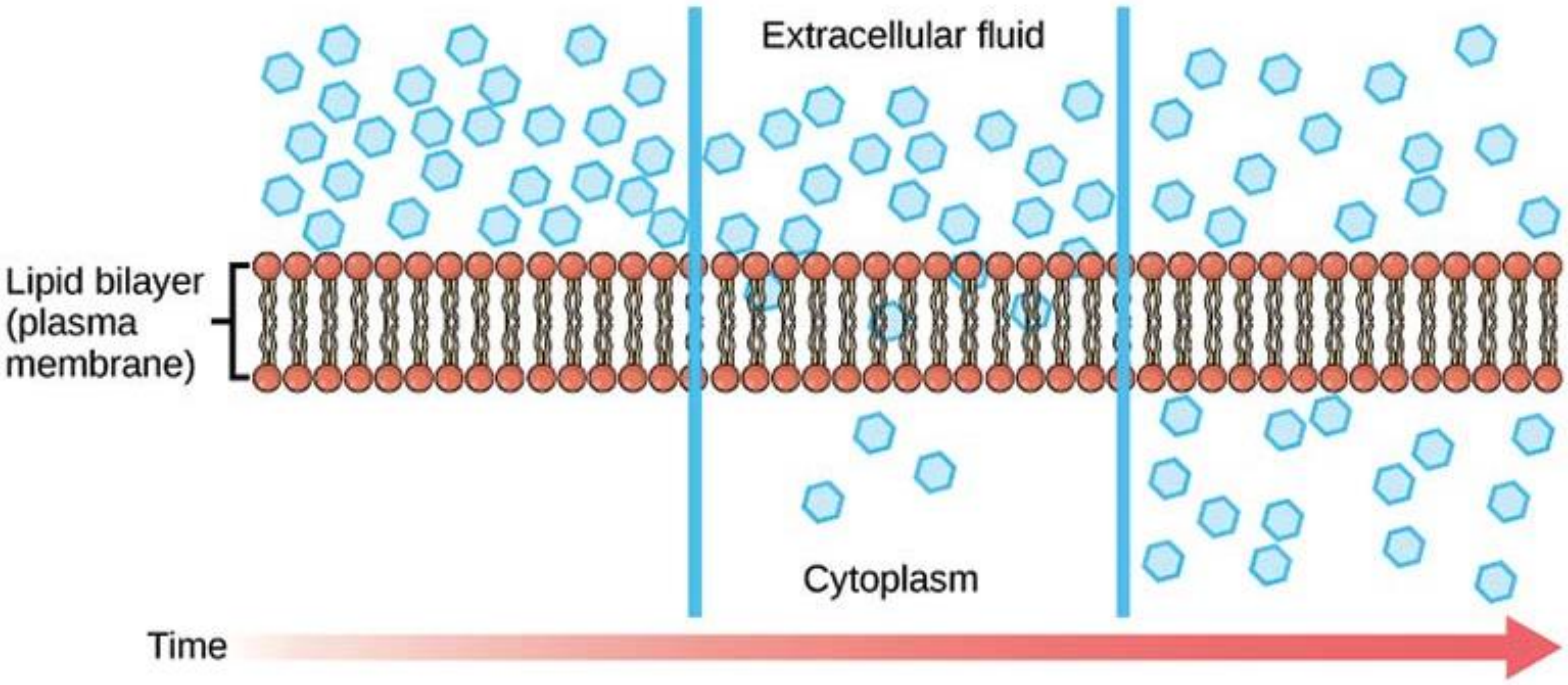
- **Physiology processes:**
  - Growth
  - Metabolic activities
  - Sensory
- **Diseases:**
  - Defective transporter: cystic fibrosis
  - Defective channel: paralysis
- **Pharmacological therapies:**
  - Hypertension: diuretics
  - Stomach ulcer: proton pump inhibitor

Fluid mosaic model of cell membrane



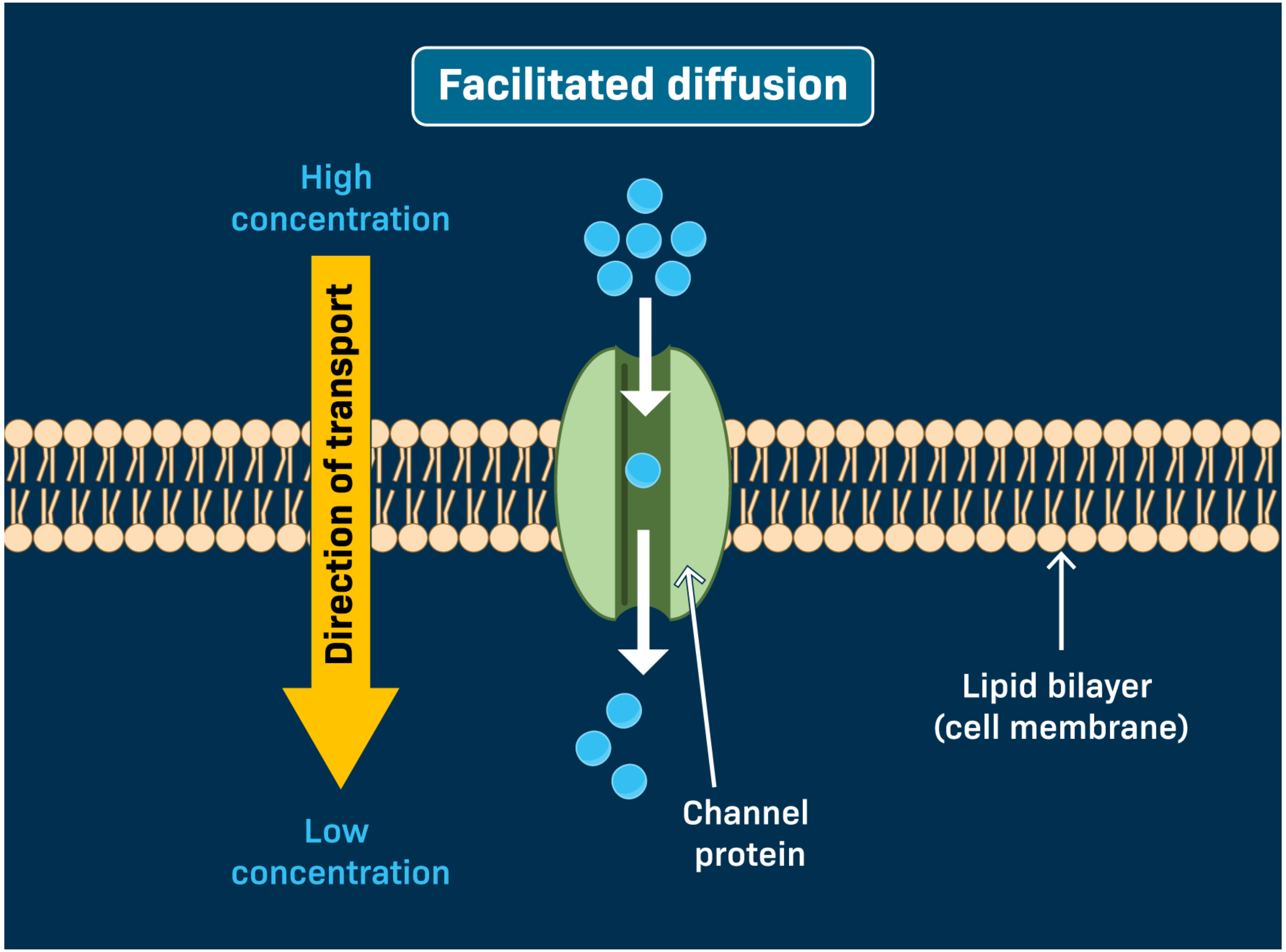
Simple diffusion

- Flux = random movement of molecule across a surface per unit time
- Net flux is determined by gradient: high to low
- Example: urea from ECF to ICF



Facilitated diffusion

- Carrier – mediated transoport



Facilitated diffusion

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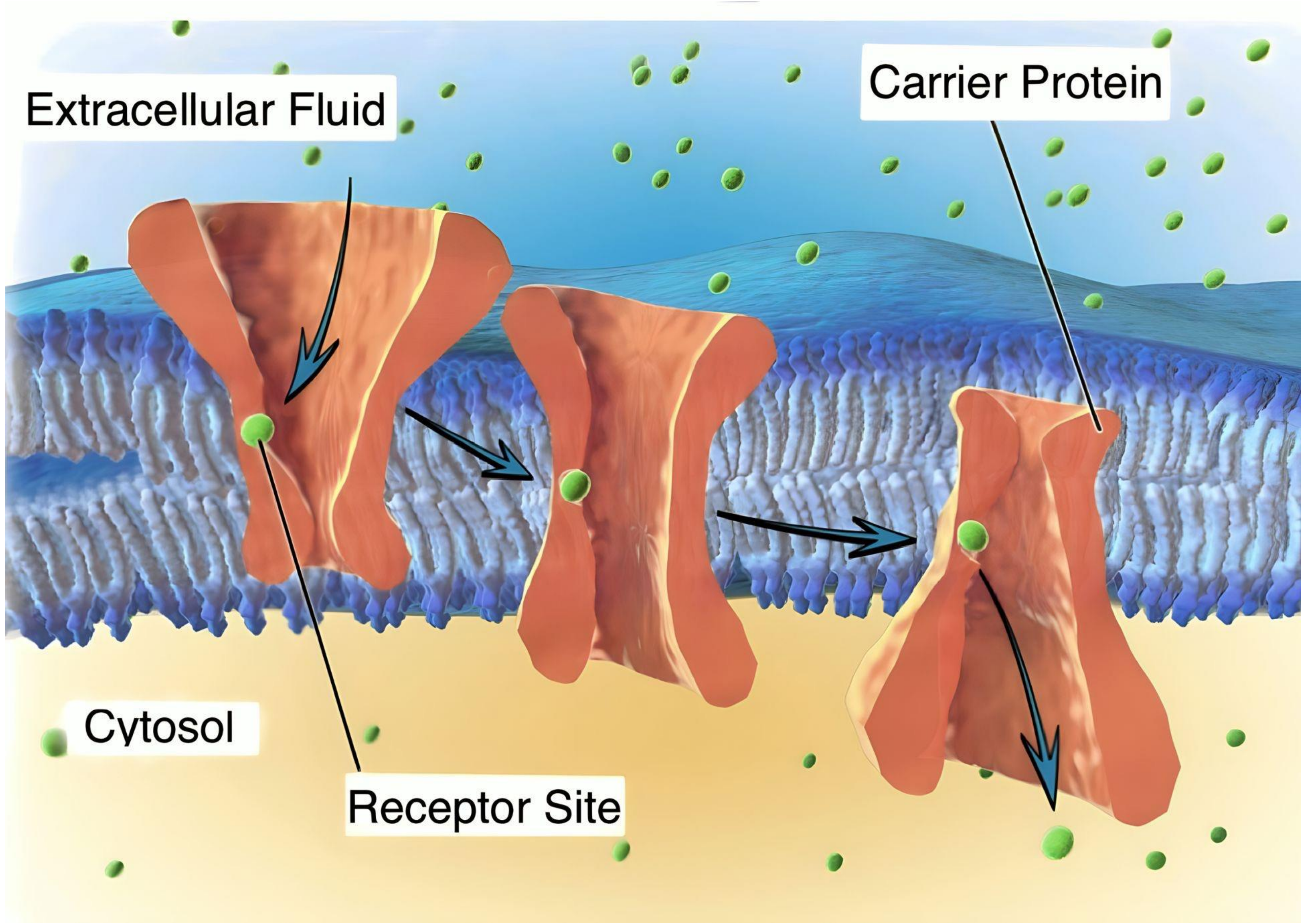
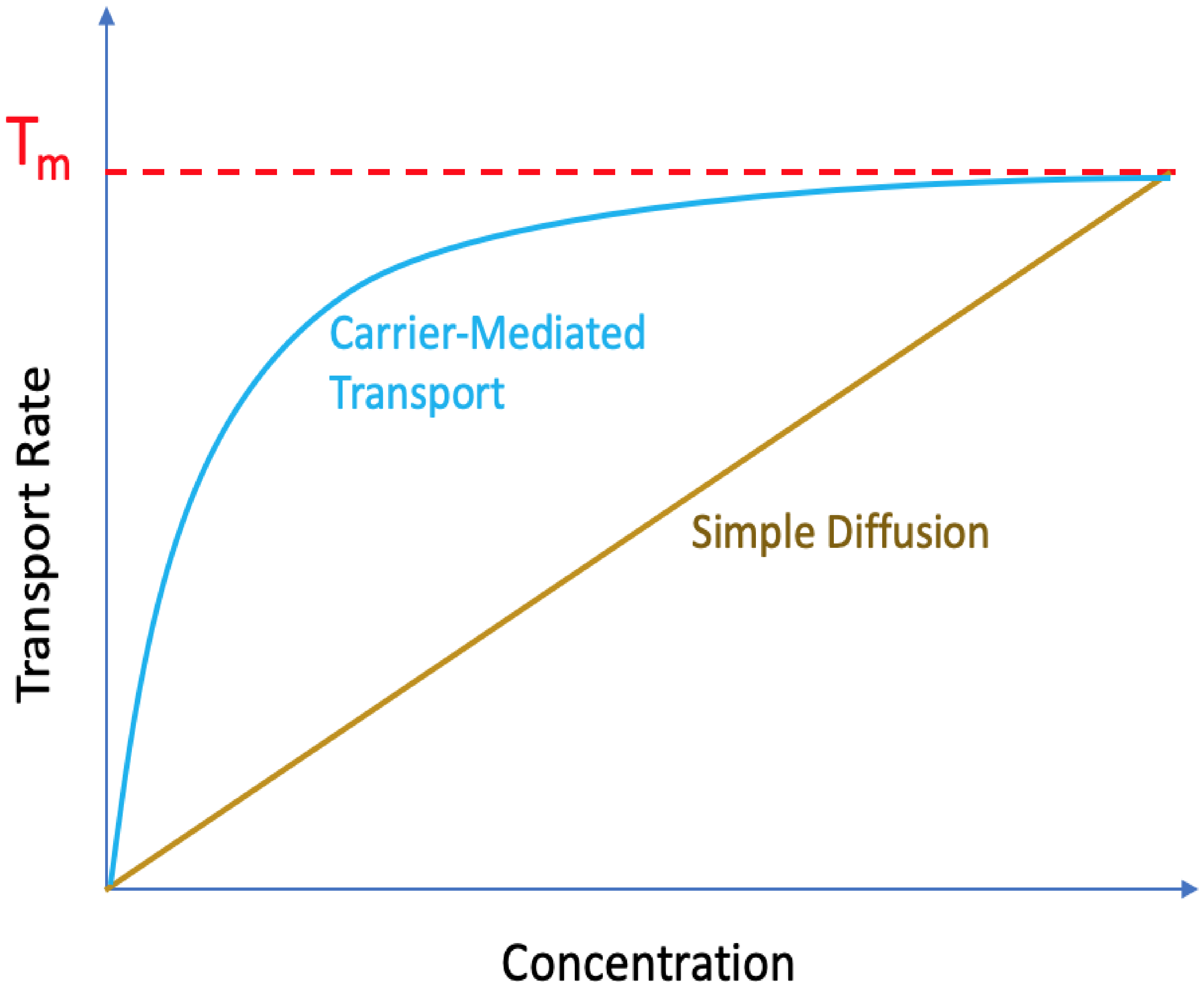
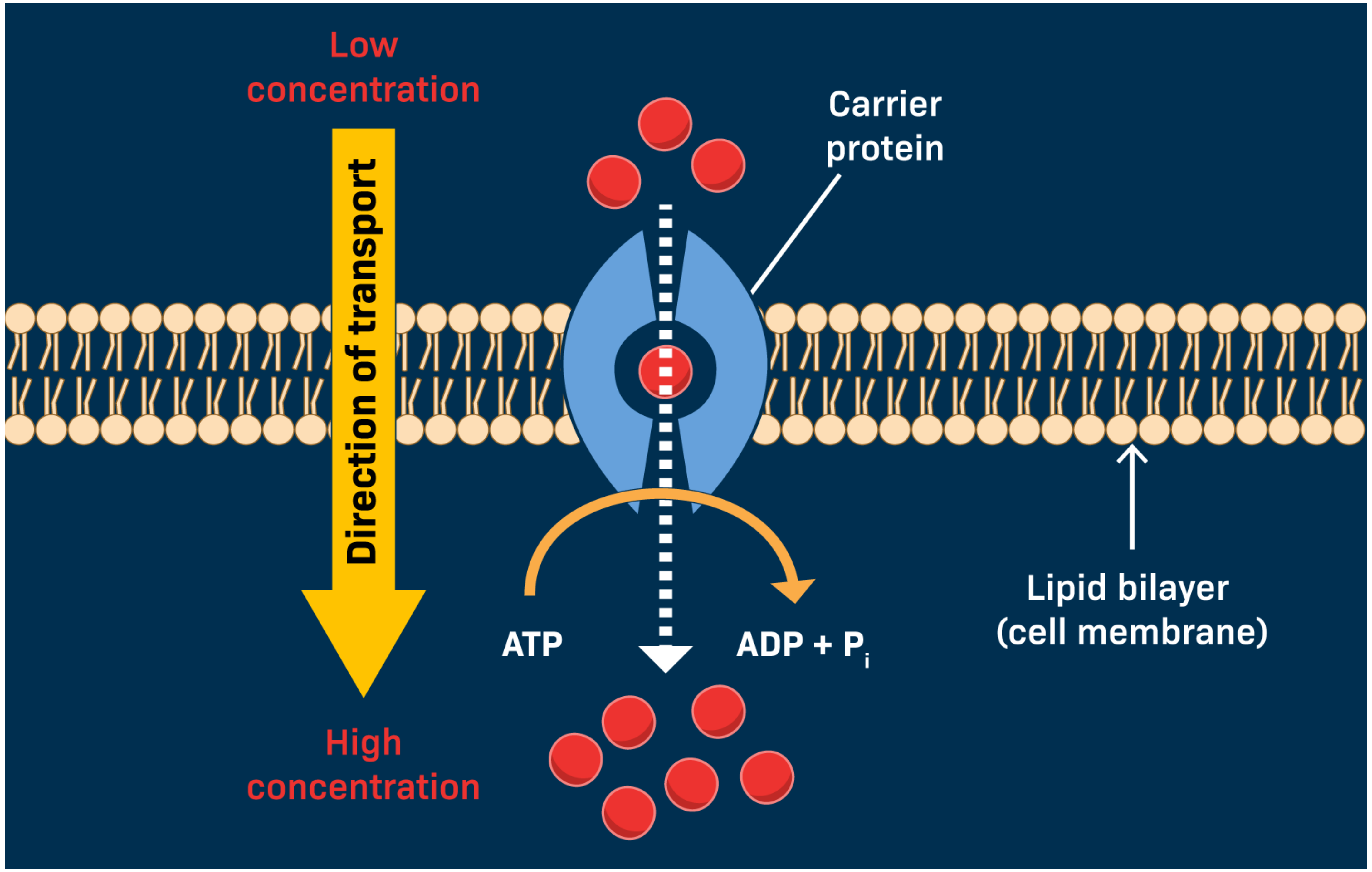




Active transport

- Against an electrochemical potential gradient (Uphill)
- Requires metabolic energy in the form of ATP



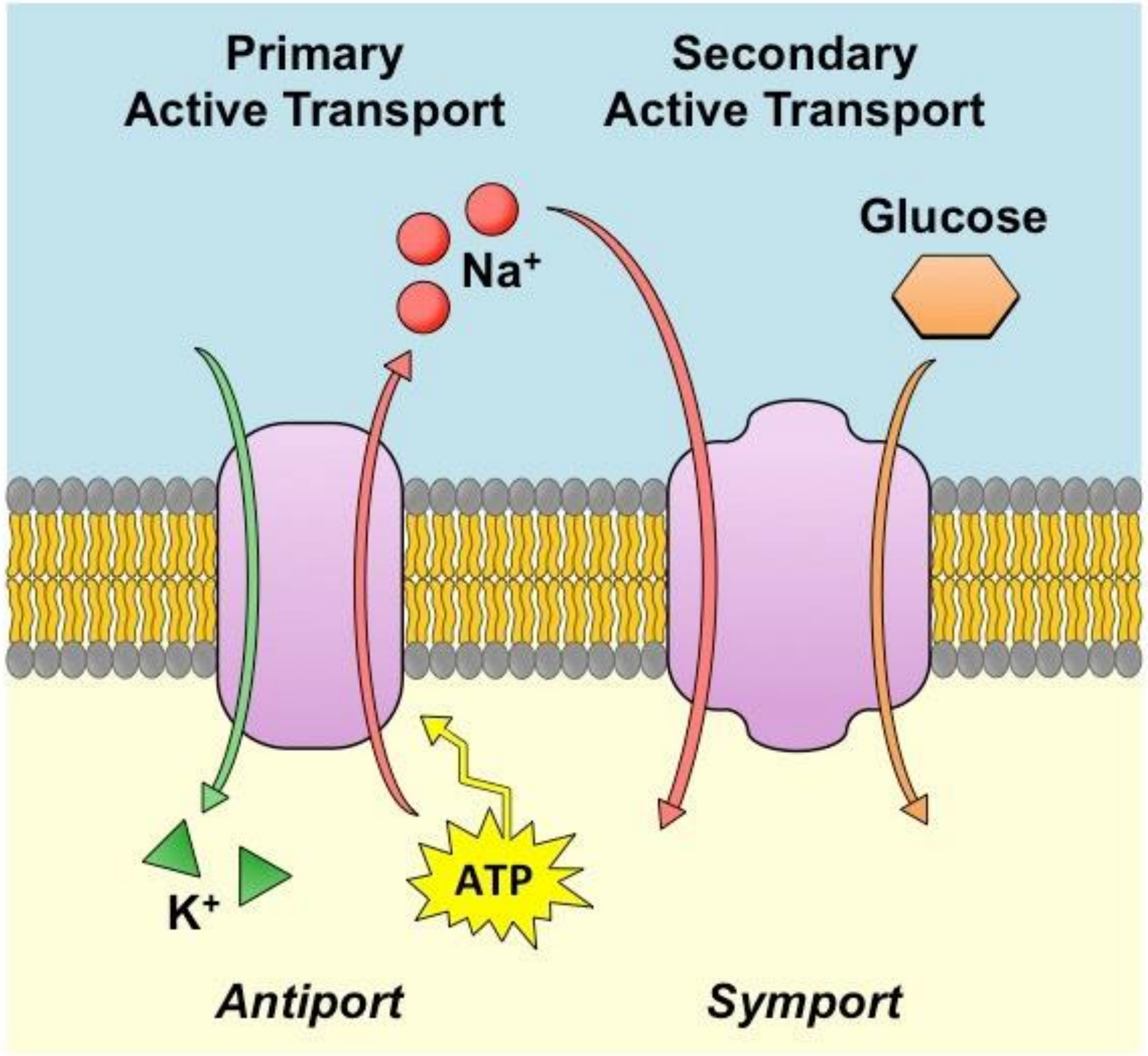
Active transport moves solute from

low **to** ***high*** **concentration**

Pums ALWAYS use energy (ATP)

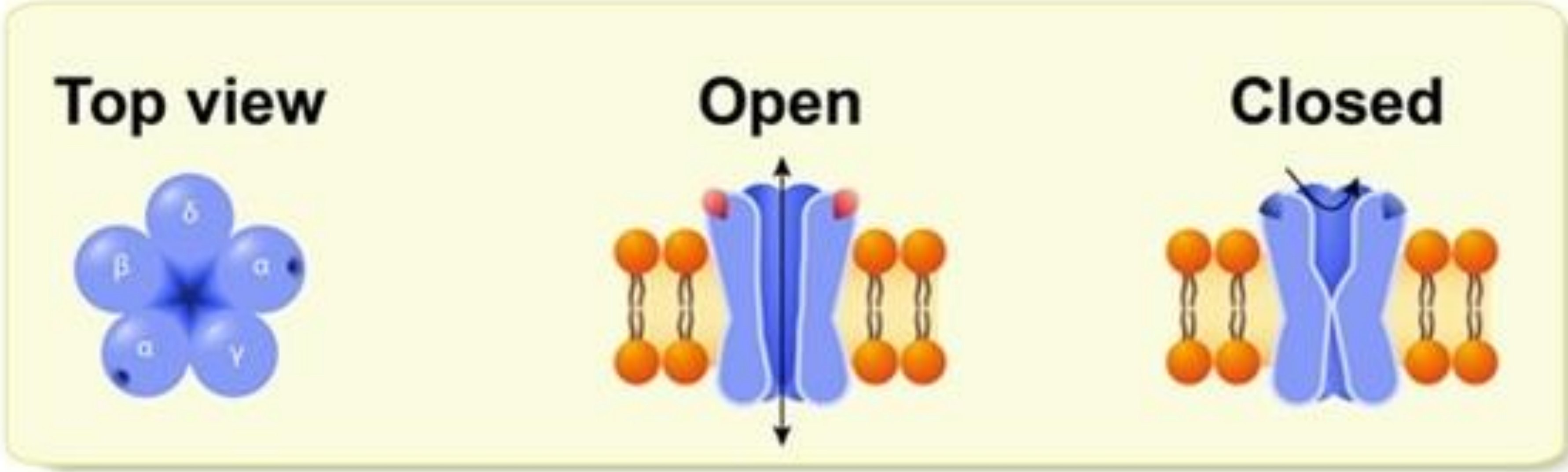
Co-transport

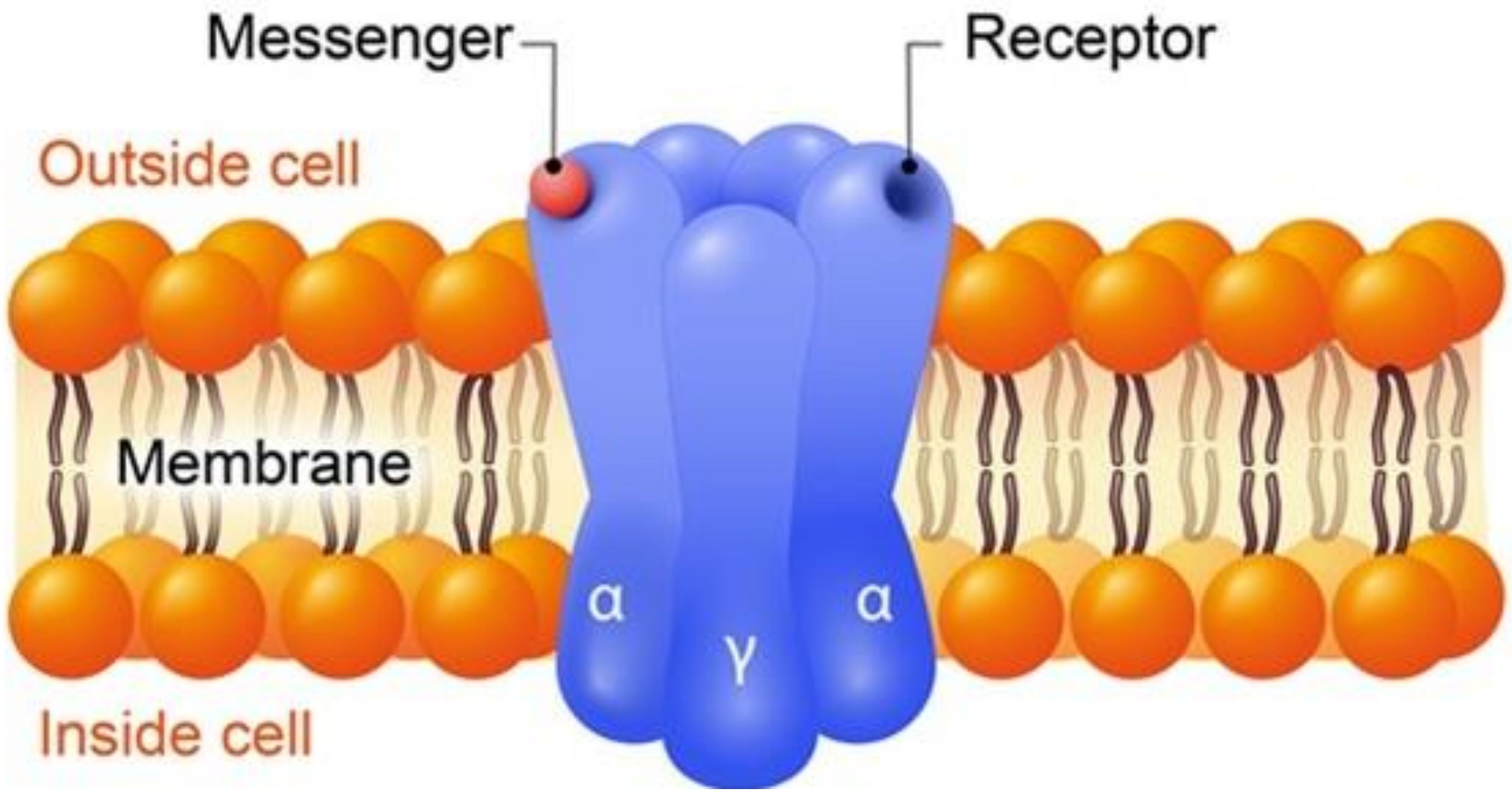
- Moves more than 1 solute at the same time
- Secondary active transport



Channels

Open = patent hole
Closed = gated

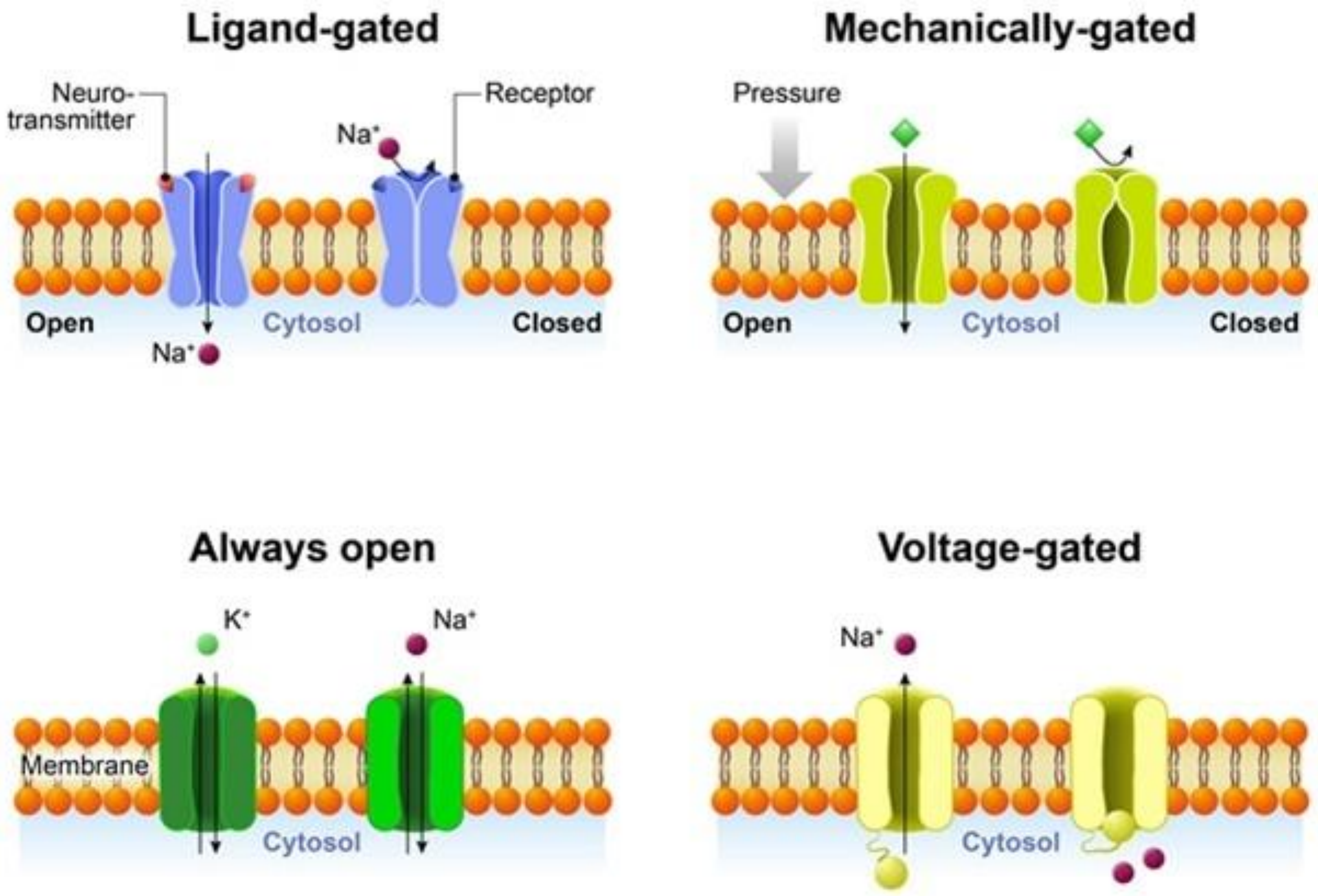


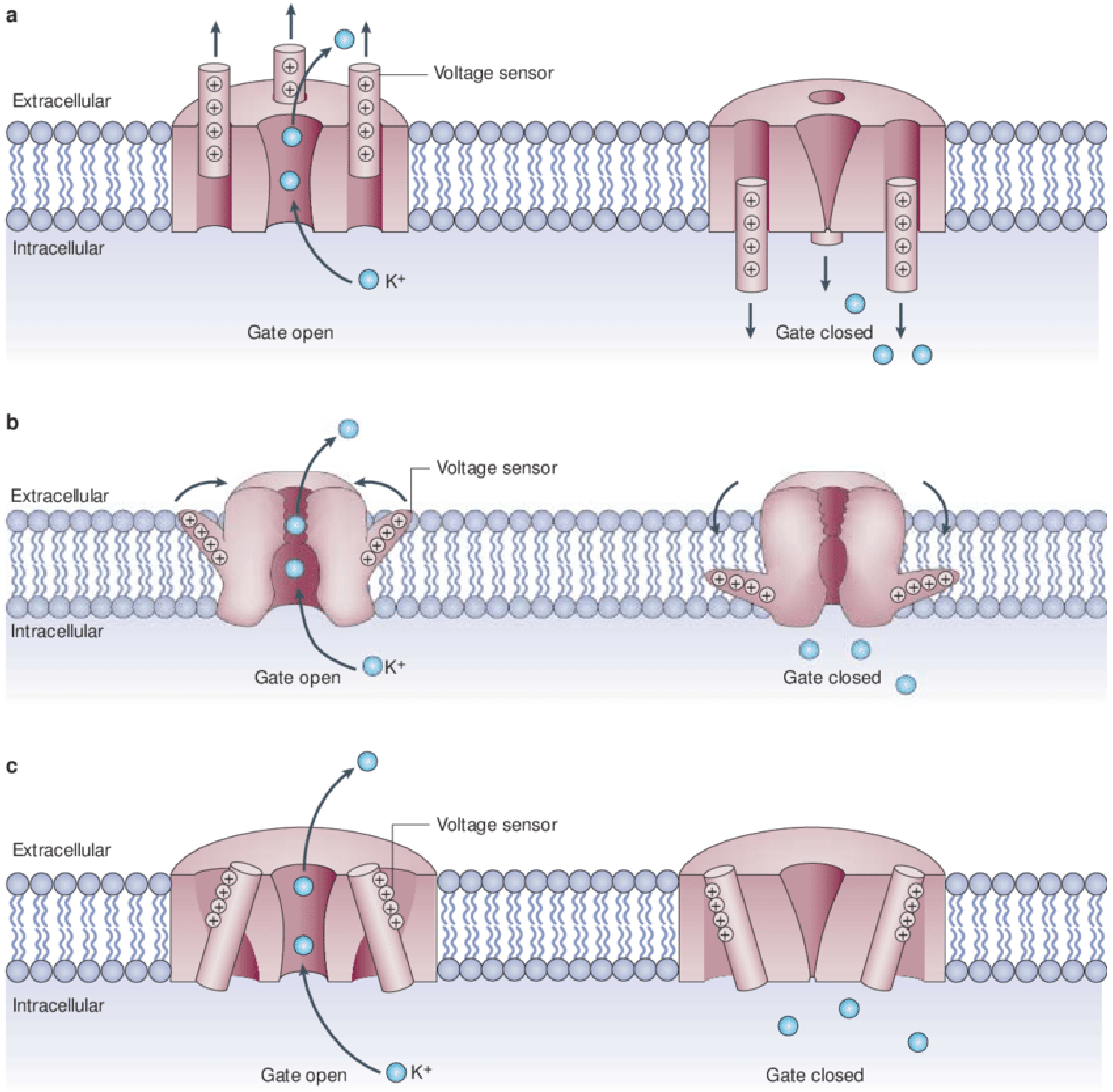


Solutes omve by diffusion from [*high*] to [*low*]

Gating of Channels

1. Ligand = requires binding of specific CHEMICAl to open
2. Voltage = requires a specific gradient of ELECTRICAL CHARGE across the membrane to open
3. Mechanical = requires specific TENSION to open



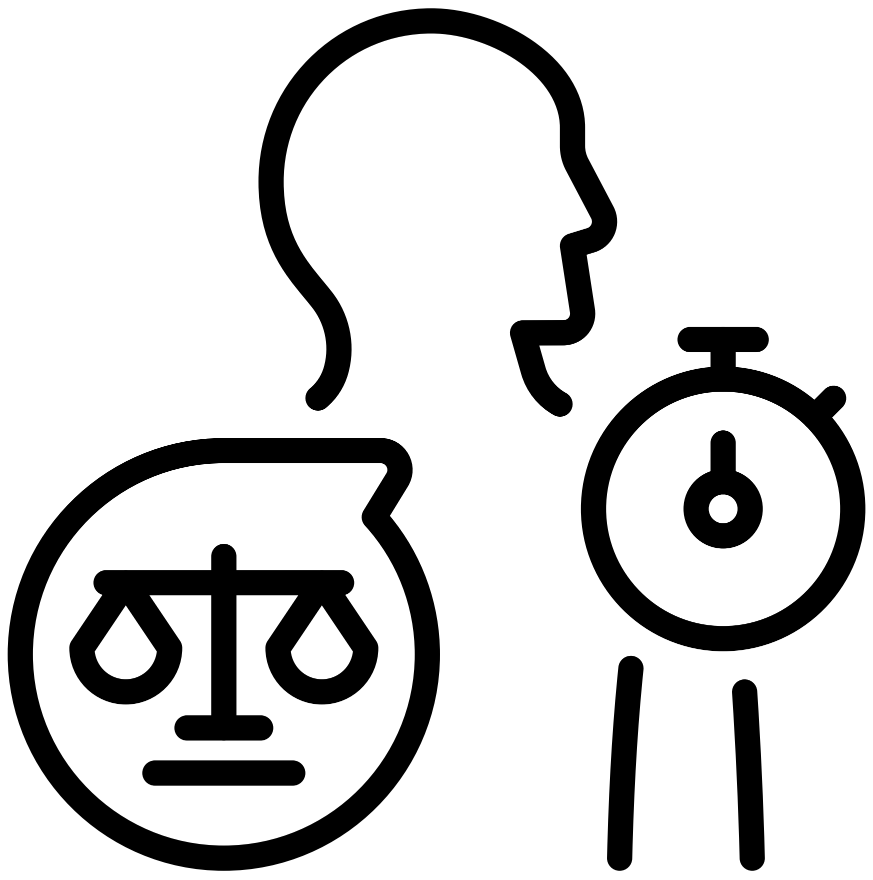


Homeostatic Imbalance

Cellular Malfunction Disruption leads to cellular damage or chronic disease states.

Clinical Outcomes Inability to restore balance results in system failure.

Repair Mechanisms Intracellular disruption may be reversible (repairable) or irreversible.



Homeostatic Imbalance

Deficiency

Cells lack vital nutrients, energy, or essential building blocks to function.

Toxicity

Cells are "poisoned" by chemical, biological, or environmental toxins.

External Factors

Environment and lifestyle choices directly impact body regulation.

Comparison of Feedback Mechanisms

	Negative Feedback	Possitive Feedback
Goal	....	....
Response	....	....
Frequency	....	....
Result	....	....